Appendix A
Selected Derivations

The technical notes are a companion track to this book, not a part of it. They hold the derivations and the technical detail stripped out of the main text so that the chapters can stay intuitive, they are maintained and published separately, and the chapters point into them with markers such as [TN14].

This appendix reproduces the two the main argument leans on hardest, so that a reader working from the book alone is not stopped by them: the quadratic-variation and Itô arguments behind Chapter 14, and the integral behind Chapter 15's formula. Each keeps its own equation prefix, so a reference to (TN15.4) means the same thing here as it does in the companion track.

A.1 ITÔ'S LEMMA: DERIVATIONS

This note supplies the arguments Chapter 14 states without proof: why $dW^2 = dt$, how Itô's lemma follows, and why a change of measure alters the drift of a diffusion but not its volatility. The level is that of a careful heuristic rather than a measure-theoretic treatment; readers wanting the latter should consult the references at the end.

A.1.1 The Quadratic Variation of a Wiener Process

Everything rests on one fact: over a fixed interval, the sum of *squared* increments of a Wiener process converges to the length of the interval, and does so with vanishing randomness.

Partition $[0, T]$ into n intervals of length $\Delta t = T/n$, and write ΔW_i for the increment over the ith. From the definition, $\Delta W_i = \varepsilon_i \sqrt{\Delta t}$ with ε_i independent standard normals. Consider

$$Q_n = \sum_{i=1}^{n} \Delta W_i^2 = \Delta t \sum_{i=1}^{n} \varepsilon_i^2 \tag{TN14.1}$$

The mean. Since $\mathbb{E}[\varepsilon_i^2] = 1$,

$$\mathbb{E}[Q_n] = \Delta t \cdot n = T \tag{TN14.2}$$

independently of n.

The variance. For a standard normal, $\mathbb{E}[\varepsilon^4] = 3$, so $\mathrm{Var}(\varepsilon_i^2) = 3 - 1 = 2$. The ε_i are independent, so

$$\mathrm{Var}(Q_n) = \Delta t^2 \cdot n \cdot 2 = 2\,\frac{T^2}{n^2} \cdot n = \frac{2T^2}{n} \tag{TN14.3}$$

As $n \to \infty$ the variance vanishes while the mean stays at T. Therefore

$$\sum_{i=1}^{n} \Delta W_i^2 \;\longrightarrow\; T \quad \text{in mean square} \tag{TN14.4}$$

The limit is not a random variable. This is the content of the informal statement $dW^2 = dt$: squared increments accumulate deterministically, at rate one per unit time.

It is worth contrasting this with the increments themselves. By the same partition, $\sum_i \Delta W_i = W(T) - W(0) \sim N(0, T)$, which remains random however fine the partition. Squaring destroys the randomness; adding does not.

Why ordinary calculus fails. Consider instead a smooth function $x(t)$ with bounded derivative. Then $\Delta x_i = O(\Delta t)$, so $\sum_i \Delta x_i^2 = O(n\,\Delta t^2) = O(\Delta t) \to 0$. For smooth functions the quadratic variation is *zero*, which is precisely why second-order terms are discarded in ordinary Taylor expansions. A Wiener path has $\Delta W_i = O(\sqrt{\Delta t})$, so its quadratic variation is $O(n\,\Delta t) = O(T)$—finite and nonzero. *The whole difference between the two calculi is the difference between $O(\Delta t)$ and $O(\sqrt{\Delta t})$ increments.*

Two corollaries follow by the same argument and are used below:

$$\sum_i \Delta W_i\,\Delta t \to 0, \qquad \sum_i \Delta t^2 \to 0 \tag{TN14.5}$$

so in the shorthand of Chapter 14, $dW\,dt = 0$ and $dt^2 = 0$, while $dW^2 = dt$ alone survives.

A.1.2 Itô's Lemma

Let S follow the Itô process

$$dS = a(S,t)\,dt + b(S,t)\,dW \tag{TN14.6}$$

and let $f(S,t)$ be twice differentiable in S and once in t. Expand f to second order:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \frac{\partial^2 f}{\partial S\,\partial t}\Delta S\,\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial t^2}\Delta t^2 + \cdots \tag{TN14.7}$$

Now substitute the discrete form of (TN14.6), namely $\Delta S = a\,\Delta t + b\,\varepsilon\sqrt{\Delta t}$, and count orders.

The squared term is

$$\Delta S^2 = a^2\Delta t^2 + 2ab\,\varepsilon\,\Delta t^{3/2} + b^2\varepsilon^2\,\Delta t \tag{TN14.8}$$

Of these three, the first two are $o(\Delta t)$ and vanish when summed, by (TN14.5). The third is $O(\Delta t)$ and, by Section A.1.1, its sum converges to $\int b^2\,dt$ with no residual randomness. Hence

$$\Delta S^2 \;\longrightarrow\; b^2\,dt \tag{TN14.9}$$

The remaining second-order terms in (TN14.7) contain $\Delta S\,\Delta t = O(\Delta t^{3/2})$ and Δt^2, both of which vanish. Substituting (TN14.9) and (TN14.6) into (TN14.7) and passing to the limit gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}a + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}b^2\right)dt + \frac{\partial f}{\partial S}\,b\,dW} \tag{TN14.10}$$

Setting $a = \mu S$ and $b = \sigma S$ recovers equation (14.8) of the main text.

The two-variable case. For $f(x,y,t)$ with $dx = a_1 dt + b_1 dW_1$, $dy = a_2 dt + b_2 dW_2$ and $dW_1 dW_2 = \rho\,dt$, the same expansion carries an additional cross term $f_{xy}\Delta x\Delta y$. By the argument of Section A.1.1 applied to the product,

$$\sum_i \Delta x_i\,\Delta y_i \;\longrightarrow\; \int b_1 b_2 \rho\,dt \tag{TN14.11}$$

which produces the extra drift term $f_{xy}b_1b_2\rho$ in equation (14.18).

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A.1.3 The Lognormal Solution

Chapter 14 asserts that geometric Brownian motion has the explicit solution used for simulation. Apply (TN14.10) to $G = \ln S$ with $a = \mu S$ and $b = \sigma S$:

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

so

$$d \ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma \, dW \tag{TN14.12}$$

The coefficients are constant, so this integrates directly:

$$\ln S_T = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma \, (W_T - W_0) \tag{TN14.13}$$

and since $W_T - W_0 \sim N(0, T)$,

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T, \quad \sigma^2 T\right) \tag{TN14.14}$$

Exponentiating (TN14.13) gives the exact simulation scheme of equation (14.6).

Moments. If $\ln X \sim N(m, s^2)$ then, completing the square in the defining integral,

$$\mathbb{E}\left[X^k\right] = \mathbb{E}\left[e^{k \ln X}\right] = e^{km + k^2 s^2/2} \tag{TN14.15}$$

Setting $k = 1$ gives $\mathbb{E}[X] = e^{m + s^2/2}$; setting $k = 2$ and subtracting the square of the mean gives the variance quoted in the main text. Applying (TN14.15) with $k = 1$ to (TN14.14),

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

confirming that the Itô correction is exactly what makes the expected price grow at μ.

A.1.4 Why the Measure Change Preserves Volatility

Chapter 14 states that moving to the risk-neutral measure changes a diffusion's drift but leaves its volatility untouched. The reason is visible from Section A.1.1 without any formal apparatus.

A change of probability measure re-weights how likely each path is. It does not alter the paths themselves, and volatility is a property of a path rather than of its probability: by (TN14.4), the quadratic variation of S along any given path is $\int \sigma^2 S^2 \, dt$, computed from the path alone. Since two equivalent measures agree on which paths are possible—they differ only in the weights—they must agree on quadratic variation, and therefore on σ.

The drift enjoys no such protection. It is defined by an *expectation*, and expectations are exactly what re-weighting changes.

The formal statement. Girsanov's theorem makes this precise. If W is a Wiener process under P and λ is a suitably regular process, then

$$W_t^Q = W_t + \int_0^t \lambda_u \, du \tag{TN14.16}$$

is a Wiener process under a measure Q equivalent to P, with the change of measure given by the density

$$\frac{dQ}{dP} = \exp\left(-\int_0^T \lambda_u \, dW_u - \frac{1}{2}\int_0^T \lambda_u^2 \, du\right) \tag{TN14.17}$$

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Substituting (TN14.16) into $dS = \mu S\,dt + \sigma S\,dW$ gives

$$dS = (\mu - \lambda\sigma)S\,dt + \sigma S\,dW^{Q}$$

so choosing $\lambda = (\mu - r)/\sigma$—the **market price of risk**—produces the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$. The volatility is unchanged, as claimed, and only the drift has moved.

The quantity λ deserves a name because it recurs: it is the excess return per unit of volatility, and Chapter 28 shows that changing numeraire amounts to changing λ, which is why the convexity, timing, and quanto adjustments of Chapter 30 all take the form of a drift correction and never a volatility correction.

Further Reading

Neftci's *An Introduction to the Mathematics of Financial Derivatives* works through the arguments above at greater length and with more attention to what is being glossed over. Baxter and Rennie's *Financial Calculus* gives a short account of Girsanov's theorem and the martingale representation theorem, and is the place to start on both. Readers wanting the measure-theoretic treatment should go to Karatzas and Shreve, or to Shreve's second volume, which is written for exactly this audience.

A.2 THE BLACK–SCHOLES–MERTON FORMULA: DERIVATIONS

Chapter 15 states the formula and explains what its terms mean. This note carries out the integral that produces it, shows the equivalence with solving the differential equation, and derives the dividend and American-approximation results the chapter uses.

A.2.1 The Risk-Neutral Integral

Under the risk-neutral measure, [TN14] established that

$$\ln S_T \sim N\!\left(m,\, s^2\right), \qquad m = \ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T, \qquad s = \sigma\sqrt{T} \tag{TN15.1}$$

and the call is the discounted expected payoff,

$$c = e^{-rT}\,\mathbb{E}^{Q}[\max(S_T - K,\,0)] = e^{-rT}\,\mathbb{E}^{Q}[(S_T - K)\mathbf{1}\{S_T > K\}] \tag{TN15.2}$$

Split it into two pieces:

$$c = e^{-rT}\underbrace{\mathbb{E}^{Q}[S_T\mathbf{1}\{S_T > K\}]}_{A} - e^{-rT}K\underbrace{Q(S_T > K)}_{B} \tag{TN15.3}$$

The second piece: B

Write $\ln S_T = m + sZ$ with Z standard normal. Then $S_T > K$ if and only if

$$Z > \frac{\ln K - m}{s}$$

so, using the symmetry $1 - N(x) = N(-x)$,

$$B = N\!\left(\frac{m - \ln K}{s}\right) = N\!\left(\frac{\ln(S_0/K) + \left(r - \sigma^2/2\right)T}{\sigma\sqrt{T}}\right) = N(d_2) \tag{TN15.4}$$

which identifies d_2 and confirms Chapter 15's statement that $N(d_2)$ is the risk-neutral probability of exercise.

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The first piece: A

Here the useful step is to complete the square. Writing $S_T = e^{m+sZ}$ and ϕ for the standard normal density,

$$A = \int_{z^*}^{\infty} e^{m+sz}\,\phi(z)\,dz = \frac{1}{\sqrt{2\pi}}\int_{z^*}^{\infty} \exp\!\left(m + sz - \frac{z^2}{2}\right) dz, \qquad z^* = \frac{\ln K - m}{s}$$

The exponent is

$$m + sz - \frac{z^2}{2} = m + \frac{s^2}{2} - \frac{(z-s)^2}{2}$$

so, substituting $y = z - s$,

$$A = e^{m+s^2/2}\,\frac{1}{\sqrt{2\pi}}\int_{z^*-s}^{\infty} e^{-y^2/2}\,dy = e^{m+s^2/2}\,N(s - z^*) \tag{TN15.5}$$

Two substitutions finish it. First, $e^{m+s^2/2} = S_0 e^{rT}$, the lognormal moment result of [TN14] and the same cancellation of $\sigma^2/2$ noted there. Second,

$$s - z^* = \frac{m - \ln K}{s} + s = d_2 + \sigma\sqrt{T} = d_1$$

so $A = S_0 e^{rT} N(d_1)$.

Assembling

Substituting A and B into (TN15.3):

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{TN15.6}$$

The put follows from put–call parity rather than a second integral, since parity holds identically.

$N(d_2)$ arose as a probability; $N(d_1)$ arose from completing the square, that is, from the shift $z \mapsto z - s$ induced by weighting each state by S_T. That shift is the content of Chapter 15's remark that $N(d_1)$ is not a risk-neutral probability: it is the probability of exercise computed under the measure in which the stock, rather than cash, is the numeraire. Chapter 28 develops that idea in general.

A.2.2 Equivalence with the Differential Equation

Chapter 15 offers two routes to (TN15.6). They agree, and here is why.

Define $V(S,t) = e^{-r(T-t)}\mathbb{E}_t^{Q}[h(S_T)]$ for any payoff h. By the law of iterated expectations, the discounted value $e^{-rt}V$ is a Q-martingale. Applying Itô's lemma to $e^{-rt}V(S,t)$ under the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$:

$$d\big(e^{-rt}V\big) = e^{-rt}\left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV\right)dt + e^{-rt}\sigma S V_S\,dW^{Q}$$

A martingale has zero drift. Setting the dt coefficient to zero gives

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV \tag{TN15.7}$$

which is equation (15.1). *The differential equation and the risk-neutral expectation are the same statement*: one says the discounted price has no drift, the other computes the expectation that follows. This equivalence is the Feynman–Kac relation, and it is why Chapter 20 can solve pricing problems either by simulating expectations or by solving partial differential equations.

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A.2.3 Dividends

A continuous yield. With a yield q, the risk-neutral drift becomes $r - q$: the asset's total return must be r, and q of it arrives as income, so the price grows at $r - q$. Repeating Section A.2.1 with $m = \ln S_0 + (r - q - \sigma^2/2)T$ changes (TN15.5) to $A = S_0 e^{(r-q)T} N(d_1)$ and leaves (TN15.4) in the same form, giving

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{TN15.8}$$

Known cash dividends, and the volatility subtlety. Chapter 15 subtracts the present value of dividends and applies the formula to $S^* = S - \mathrm{PV}(D)$. The justification is that S^* is the risky component: S equals a certain part, $\mathrm{PV}(D)$, plus an uncertain part which is assumed lognormal.

The subtlety the chapter flags can be made precise. If S^* has volatility σ^* and $S = S^* + \mathrm{PV}(D)$, then the volatilities of the two are related by

$$\sigma^* \approx \sigma \frac{S}{S^*} = \sigma \left(1 - \frac{\mathrm{PV}(D)}{S}\right)^{-1} \tag{TN15.9}$$

because the same absolute risk is carried by a smaller base. For a stock yielding 6% with a two-year option, $\mathrm{PV}(D)/S \approx 0.113$, so σ^* exceeds the observed σ by roughly 13%, which is not negligible. Using the unadjusted volatility understates the option.

A.2.4 Black's Approximation

Chapter 15 states the rule; here is why it takes that form.

Section 11.10 established that an American call on a stock paying a single dividend at t_1 can be optimally exercised only at two moments: immediately before t_1, or at expiration T. Exercise elsewhere is dominated, because the interest saved by deferring the strike payment exceeds any benefit until the dividend is imminent.

The value of an American call is the maximum over admissible exercise policies. Restricting attention to those two:

$$C \approx \max\left(\underbrace{c(S_0 - \mathrm{PV}(D),\, K,\, T)}_{\text{hold to } T},\ \underbrace{c(S_0,\, K,\, t_1)}_{\text{exercise at } t_1} \right) \tag{TN15.10}$$

The first term treats the option as European to T on the dividend-adjusted price. The second treats it as European to t_1 with no adjustment, since a holder exercising then captures the dividend and the adjustment does not apply.

The approximation *understates*, and the reason is now visible: it evaluates a fixed choice between two policies made today, whereas the true American holder decides at t_1 with knowledge of S_{t_1}. Being able to choose after observing is worth something, so

$$C \geq \max(\cdot, \cdot)$$

with equality only in the degenerate cases where one policy dominates in every state. The error is largest when the two terms are close, which is exactly when the exercise decision is genuinely uncertain.

Further Reading

Black and Scholes (1973) derive the formula through a hedged portfolio and the heat equation; Merton (1973) uses a self-financing replicating portfolio and also handles dividends and the American call. Cox and Ross (1976) is where the risk-neutral integral of Section A.2.1 first

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appears as a general method. For the Feynman–Kac equivalence of Section A.2.2, Baxter and Rennie is the short route and Shreve's second volume the measure-theoretic one.

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Appendix B
Major Derivatives Exchanges

Chapter 2 described what an exchange does; this appendix says who they are. The list covers venues that matter either because of their volume or because they own a benchmark that the rest of the market prices against, and for each it names the group that owns it, where it sits, and what actually trades there.

Two warnings belong before the tables.

Ownership and product lists move. Exchanges are acquired, renamed, and delisted; contracts are launched and retired. What follows reflects the position in early 2026, and any specific claim here should be checked against the venue's own contract specifications before it is relied on for a trade. *The structure is more durable than the details*, which is the reason the appendix closes with Section B.8 rather than with a table.

And there is no single ranking. Section B.1 explains why the two ways of measuring an exchange's size disagree by orders of magnitude, and why almost every published league table is answering a question the reader did not ask.

B.1 HOW TO READ THE TABLES

The two standard volume measures do not rank the same way, and the gap between them is not small.

Contracts traded counts tickets. It is the figure the Futures Industry Association publishes and the one exchanges quote in press releases. *It is also almost meaningless across venues*, because contract sizes differ by factors of thousands: an Indian index option on a lot worth a few thousand dollars and a Eurex Bund futures contract on €100,000 of notional each count as one.

Notional traded counts economic exposure. It ranks quite differently, and it is the figure that matters for anything in Chapters 27 and 39, where exposure and capital are the subject.

The consequence is plain. By contracts, the largest derivatives exchange in the world is in India; by notional it is in Chicago, and the two are not close in either direction. A reader who sees "largest exchange" in a headline should establish which measure is meant before drawing any inference about where risk actually sits.

Two further cautions apply to any of these figures.

Open interest is the better measure of what is held. Volume measures turnover, and a venue dominated by same-day expirations (Section 23.1's 0DTE flow, or the Indian weekly options market) generates enormous volume against small open interest. *Turnover and exposure are different questions.*

And exchange-traded volume is not the whole market. The largest interest rate and credit markets in this book are over-the-counter and clear through the houses of Section 2.4 rather than trading on any exchange listed here. *A directory of exchanges is a map of the listed market, not of derivatives.*

B.2 NORTH AMERICA

Exchange	Group	Principal derivatives
CME	CME Group	SOFR and Fed funds futures, E-mini S&P 500 and Nasdaq-100, FX futures, live cattle
CBOT	CME Group	US Treasury futures across the curve, corn, soybeans, wheat
NYMEX	CME Group	WTI crude, Henry Hub natural gas, refined products
COMEX	CME Group	Gold, silver, copper
Cboe Options	Cboe Global Markets	SPX options, VIX options, listed equity options
Cboe Futures (CFE)	Cboe Global Markets	VIX futures
ICE Futures U.S.	ICE	Sugar, coffee, cocoa, cotton, Russell 2000, USD index
Nasdaq PHLX, ISE, GEMX, MRX	Nasdaq	Listed equity and index options
MIAX Options, Pearl, Emerald, Sapphire	Miami International Holdings	Listed equity options
MGEX	Miami International Holdings	Hard red spring wheat
Montréal Exchange	TMX Group	Three-month CORRA futures, Canadian government bond futures, S&P/TSX 60 futures

The concentration is the point. Four of the eleven venues above belong to CME Group, and between them they carry the US benchmark contract in interest rates, equity indices, energy, and metals. *No other jurisdiction has anything comparable*, and Section B.8 returns to what that means for the clearing houses behind them.

One entry records a transition this book has already described. Montréal's BAX contract referenced CDOR and was retired with it; the three-month CORRA futures contract is its replacement, and Section 4.3's account of the benchmark transition is what that line in the table is about.

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B.3 EUROPE

Exchange	Group	Principal derivatives
Eurex	Deutsche Börse	Bund, Bobl, Schatz and Buxl futures, Euro Stoxx 50 futures and options, single-stock options, €STR and Euribor futures
ICE Futures Europe	ICE	Brent crude, gasoil, SONIA and Euribor futures, FTSE 100
ICE Endex	ICE	Dutch TTF natural gas, the European gas benchmark
London Metal Exchange	HKEX	Copper, aluminum, zinc, nickel, lead, tin, on the prompt-date structure
Euronext	Euronext	CAC 40, AEX, FTSE MIB, milling wheat, rapeseed, corn
EEX	Deutsche Börse	German and French power, EU emission allowances, gas
Nasdaq Nordic	Nasdaq	OMXS30 futures and options, Nordic equity derivatives
Borsa İstanbul (VIOP)	Borsa İstanbul	Equity index and single-stock futures, currency
Moscow Exchange	MOEX	Domestic index, FX, and commodity contracts; largely closed to international participants since 2022

Europe's map is drawn differently from America's, and the difference is the subject of Section B.8. Eurex clears its own contracts; most European over-the-counter interest rate risk clears at LCH, which is owned by neither an exchange group nor its members in the way a vertical silo is.

Two entries carry episodes from Chapter 40. The London Metal Exchange's nickel contract is where the March 2022 suspension and cancellation of trades happened, and the Dutch TTF gas contract is where the 2022 margin spiral of Section 40.11 ran. *Both are cases where the venue itself became the risk*, which is why they are named here rather than left to a footnote.

B.4 ASIA-PACIFIC

Exchange	Group	Principal derivatives
National Stock Exchange	NSE India	Nifty 50 and Bank Nifty index options and futures; the largest venue in the world by contracts traded
BSE	BSE India	Sensex and Bankex index options
MCX	MCX India	Gold, silver, crude oil, natural gas
Osaka Exchange	JPX	Nikkei 225 futures and options, TOPIX, JGB futures
TOCOM	JPX	Energy, rubber, precious metals
Korea Exchange	KRX	KOSPI 200 options and futures, three-year KTB futures
HKEX	HKEX	Hang Seng and HSCEI futures and options, MSCI contracts, USD/CNH futures
Singapore Exchange	SGX	FTSE China A50, Nikkei 225, iron ore, rubber, FX futures
TAIFEX	TAIFEX	TAIEX futures and options
ASX	ASX	SPI 200, three- and ten-year Treasury bond futures, 90-day bank bill
Bursa Malaysia Derivatives	Bursa Malaysia	Crude palm oil (FCPO), the global benchmark

The Indian entry deserves the qualification the table cannot carry. NSE has led the world

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by contract count for several years, on a market dominated by weekly index options with very small lot sizes, which is Section B.1's warning in its purest form. *Regulatory measures from late 2024* (larger lot sizes, and a limit of one weekly expiration per exchange) were introduced precisely because the authorities concluded that retail turnover of that shape was not serving the participants generating it. Volumes fell sharply and the ranking survived, which tells you how large the gap to second place was.

And one entry records a contract that moved countries. SGX's Nifty futures, long the offshore venue for Indian index exposure, transitioned in 2023 to NSE's international arm at GIFT City. *An offshore benchmark returning onshore is unusual*, and it is the mirror image of the Chinese pattern described next.

B.5 MAINLAND CHINA

China's exchanges are treated separately because the market is structured differently: six futures exchanges, each with a defined product remit, under a single regulator, with foreign access granted contract by contract rather than venue by venue.

Exchange	Location	Principal derivatives
Shanghai Futures Exchange (SHFE)	Shanghai	Copper, aluminum, zinc, nickel, rebar, hot-rolled coil, gold, silver, rubber
Shanghai International Energy Exchange (INE)	Shanghai	RMB-denominated crude oil, TSR20 rubber, low-sulfur fuel oil; open to overseas investors
Dalian Commodity Exchange (DCE)	Dalian	Iron ore, soybean meal and oil, palm olein, corn, coking coal, coke, LLDPE, LPG
Zhengzhou Commodity Exchange (ZCE)	Zhengzhou	PTA, methanol, cotton, sugar, rapeseed meal and oil, urea, soda ash, glass
China Financial Futures Exchange (CFFEX)	Shanghai	CSI 300, CSI 500, SSE 50 and CSI 1000 index futures and options; 2-, 5-, 10- and 30-year government bond futures
Guangzhou Futures Exchange (GFEX)	Guangzhou	Industrial silicon, lithium carbonate
Shanghai and Shenzhen Stock Exchanges	Shanghai, Shenzhen	ETF options on broad-based indices

The division of labor is administrative rather than historical, which is why the product split is so clean: ferrous and energy in Dalian, chemicals and soft commodities in Zhengzhou, non-ferrous metals in Shanghai, all financial contracts at CFFEX, and the new-energy complex at Guangzhou, which was created in 2021 for exactly that purpose. *No Western exchange group was assembled that tidily*, because none of them was assembled at all—they merged into their present shape.

Two features matter for anyone pricing these contracts. The exchanges impose daily price limits and position limits far more actively than their Western counterparts; and the government bond futures at CFFEX use a 3% notional coupon against the 6% of the CME's Treasury contracts, which changes which bond is cheapest to deliver.

Access is by contract, not by venue. INE's crude contract was built from the outset for overseas participation, and specific DCE and ZCE contracts have been opened since; most of the rest remain domestic. *Hong Kong is the offshore complement*, which is where the USD/CNH futures in Section B.4's table fit.

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B.6 LATIN AMERICA, THE MIDDLE EAST, AND AFRICA

Exchange	Location	Principal derivatives
B3	São Paulo	DI interest rate futures, Ibovespa futures and options, USD/BRL futures, single-stock options
MexDer	Mexico City	TIIE interest rate futures, IPC index, USD/MXN
Gulf Mercantile Exchange	Dubai	Oman crude, the principal East-of-Suez sour benchmark
ICE Futures Abu Dhabi	Abu Dhabi	Murban crude
Saudi Exchange	Riyadh	Index and single-stock futures
Johannesburg Stock Exchange	Johannesburg	Equity index and single-stock futures, currency, white and yellow maize, wheat

B3 is larger than its profile suggests. Its DI contract, a futures contract on the domestic interbank rate, trades in volumes that place it among the world's most active interest rate contracts by count, and Brazilian hedgers use it in place of the swap market that dominates elsewhere. *Where a listed contract is liquid enough, it displaces the over-the-counter equivalent*, which is the observation Chapter 2 makes in the abstract and Brazil demonstrates.

The two Gulf entries exist because of a gap in the crude benchmark map. Brent and WTI are light and sweet; much of what actually moves from the Gulf to Asia is neither, and both Oman and Murban were listed to give that barrel a transparent price. Section 36.9's discussion of benchmark grades is what these contracts are for.

B.7 CRYPTO VENUES

Chapter 37 treats these instruments; what belongs here is the distinction between the venues that trade them, because it is sharper than in any other asset class.

Regulated futures exchanges list bitcoin and ether futures and options under the same rules as any other contract. CME is much the largest of these, and its contracts are the ones institutional participants can hold without a separate custody and counterparty framework.

Offshore crypto-native venues carry most of the volume. They list the perpetual futures of Section 37.3 (a contract with no delivery date, held in line with spot by the funding-rate mechanism), and they clear, custody, and match on the same balance sheet, which is the vertical integration Section 37.10 identifies as the structural risk. *The largest of them trade more notional than the regulated venues by a wide margin*, and Deribit has long been the dominant venue for crypto options specifically.

The distinction is not about size but about what fails when something fails. A CME position is protected by the clearing arrangements of Chapter 2 and the segregation rules of Chapter 8. *On a venue that is simultaneously exchange, clearing house, and custodian, those three protections are one balance sheet*, and Section 37.10's account of what that has cost is the reason the appendix separates them.

B.8 WHAT THE MAP SHOWS

Four structural facts survive the details, and they are the reason to read a directory of this kind at all.

The listed market is concentrated in a small number of groups. CME Group, ICE, Cboe, Nasdaq, Deutsche Börse, HKEX, JPX, NSE, and B3 account for the large majority of exchange-traded derivatives volume worldwide. *Competition happens between groups, not between the venues inside them*, which is why a group's several exchanges rarely list the same contract against each other.

Liquidity does not migrate. A benchmark contract is a network good: it is liquid because

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everyone uses it, and everyone uses it because it is liquid. *Competing exchanges have repeatedly listed identical contracts against an incumbent and repeatedly failed*, which is why the Bund is at Eurex and Treasuries are at CBOT decades after either could in principle have moved.

The vertical silo is the dominant model outside Europe. An exchange that owns its clearing house captures the margin as well as the fee and, more consequentially, makes positions non-fungible with any competitor's. Europe's separation of LCH from the trading venues is the main exception, and Section 2.10's discussion of clearing concentration is where the consequences are worked out.

And the exchange is not where most derivatives risk sits. Interest rate swaps, credit default swaps, and the structured products of Chapter 35 are over-the-counter instruments; they clear centrally, but they do not trade on a venue in this appendix. *The exchanges hold the price discovery and the retail flow; the dealers and the clearing houses hold the exposure.* Anyone reading this list as a map of where the risk is has read it as the opposite of what it is.

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Appendix C
Glossary

Definitions are working ones: what the term means in this book and where it is developed, not what a dictionary would say. *A chapter number in parentheses is where to go*, and where a term is defined precisely in one place and used loosely elsewhere, the pointer is to the precise one.

A

Accreting swap A swap whose notional steps up over time, matching a drawdown (33).

Accrued interest The part of a bond's next coupon earned since the last coupon date, measured on the bond's day count and paid by the buyer to the seller on top of the quoted price (6).

Additional valuation adjustment (AVA) The gap between a position's fair value and a prudent value it could be realized at with 90% confidence, deducted directly from common equity under the EU and UK prudent valuation rules. Nine categories, among them market price uncertainty, close-out cost and model risk (39).

Add-on In SA-CCR, the supervisory stand-in for how far an asset class's positions can move over the horizon that matters: for each hedging set, the supervisory factor times the effective notional, summed over the class's hedging sets. Scaled by the multiplier, the add-ons make up potential future exposure (27).

Adjoint algorithmic differentiation (AAD) Computing every sensitivity of a simulated price by running the chain rule backward through the stored pricing calculation, from the price to each input, in a fixed multiple (typically three to five times) of the cost of the price itself, whatever the number of inputs. What makes a model with many calibration inputs viable on a desk (9, 20, 32).

Admissible strategy A trading strategy whose value never falls below a fixed floor. The restriction that makes the first fundamental theorem true in continuous time, by ruling out doubling strategies (TN28).

Affine model A short-rate model in which $\ln P(t,T)$ is affine in the short rate (linear plus a constant), so bond prices have a closed form and every yield on the curve is an affine function of one state variable. Vasicek, CIR and Hull–White are all in this class (31).

American Monte Carlo Simulation that handles early exercise, or the future revaluation of a portfolio, by estimating each path's continuation value with a least-squares regression across paths: the Longstaff–Schwartz method. Exposure systems borrow the regression to avoid revaluing every trade exactly on every path (9, 20).

American option An option exercisable at any time up to expiration (10, 20).

Amortizing swap A swap whose notional steps down, matching a loan being repaid. Pricing it at its weighted average life is a trap (33).

Annuity factor The present value of one unit paid on each fixed-leg payment date, weighted by the accrual factors, so one basis point of fixed rate on unit notional is worth 0.0001 times it; the natural numeraire for swaptions (7, 29, 30).

Annuity mapping function In CMS pricing, the conditional expectation $\alpha(s)$ of the reciprocal

of the annuity given that the swap rate finishes at s. It gathers every unknown about the curve's joint law into one function, so a CMS payoff becomes an integral against the swap rate's marginal law; freezing the annuity makes it constant, and a linear terminal swap rate model makes it affine (28).

Antithetic variates Pairing every simulated path with its mirror image, built from the same random shocks with their signs reversed, and averaging the two payoffs. The pair is negatively correlated, so the variance of the estimate falls at almost no cost (20).

Arbitrage A trading strategy that costs nothing to set up, can never lose money, and makes money in at least one state of the world. Its absence is the assumption that prices everything in this book (5, 11).

Arrow–Debreu price The value today of one unit paid at one node of a lattice and nothing anywhere else; the state price of Chapter 13 written on a tree. Summing them across a level returns that maturity's discount factor, which is what the Hull–White construction solves against (13, 31).

Asian option An option on the average price over a period. Cheaper than a vanilla because an average has less variance than an endpoint (34).

Assessment rights A CCP's right, near the bottom of its default waterfall, to demand further contributions from surviving clearing members once the default fund is exhausted. It allocates a loss rather than creating resources, and it transmits stress to firms already struggling (2).

Asset-swap spread The spread over SOFR received when a fixed-rate bond is bought at par and its coupons are swapped to floating; the measure of a bond's credit spread that a CDS–bond basis trade earns (24).

At-the-money Strike equal to the current spot or forward; the two conventions differ and the difference matters for quoting (19).

At-the-money forward A strike equal to the forward price, at which a European call and a European put have equal value by put–call parity. The usual reference point for quoting options, and the strike of the straddle used to trade volatility without a directional view (11, 19).

Attachment point The loss level at which a tranche begins to absorb losses (25, 26).

Autocallable A structured note that redeems early, repaying its face plus the coupons accrued, if the underlying is at or above a trigger level on an observation date. If it is never called, the buyer is short a put that knocks in below a barrier (35).

Automated market maker A venue that quotes from a formula over a pooled inventory rather than from an order book (37).

B

Backtesting / exception Checking a risk measure against realized losses by counting exceptions, the days on which the loss exceeds the VaR; at 99% over 250 days a correct model expects 2.5. Counting is weak evidence: anything from one to six exceptions is consistent with a correct model, zero is a rejection too, and the count cannot see whether exceptions cluster. The count sets the capital multiplier under internal models, and it is VaR that is backtested, because expected shortfall cannot be tested by counting (38, 40).

Backward induction Valuing a derivative on a tree by working back from expiration one step at a time, each node's value being the discounted risk-neutral expectation of the values at the nodes that follow it. For an American option each node also compares that value with immediate exercise (13).

Backward-looking caplet A caplet on a rate compounded in arrears over its accrual period, so its payoff is known only at the period's end. The rate keeps accumulating variance through

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the period, so Black's formula applies with $\sigma^2 T_1$ replaced by $\sigma^2(T_1 + \tau/3)$, and the caplet is worth more than its forward-looking counterpart (29).

Backwardation A forward curve sloping downward. Distinct from Keynes's *normal* backwardation, which compares the futures price with the expected spot (3, 5).

Banking Keeping an emissions allowance not surrendered this year for use in a later one. Because banking is costless and unlimited, the forward cannot exceed spot carried at the financing rate, and that upper bound binds tightly; because an allowance cannot be borrowed, the forward can sit below carry, and the residual measures doubt about the scheme. The limiting case of a storable commodity (36).

Barrier option An option extinguished or created when the underlying touches a level. The largest exotic market, and among the instruments with the most model risk, with cliquets and forward-start options (34).

Base correlation A quoting convention that removes the inconsistency of implied correlation across tranches without removing what caused it (25).

Basel framework The international bank capital standards agreed by the Basel Committee on Banking Supervision and implemented, with national variations, by each jurisdiction's regulator. It sets what a derivatives position costs in capital, and therefore what it costs to hold (9, 27, 40).

Basis The spot price minus the futures price, which converges to zero at delivery. What a hedger is left with when the hedge is lifted before delivery (2, 3). In credit, the CDS spread minus the bond's spread, which the arbitrage of buying the bond and buying protection says should be zero and which persists (24).

Basis swap A swap exchanging two floating rates, pricing the spread between them (7, 33).

Basket default swap Credit protection on a small basket of issuers, typically five to ten, that pays on the nth default and then terminates. The first-to-default basket is the common case; to first order its value depends on the issuers' hazard rates rather than on recovery, and correlation makes it cheaper than the sum of its parts, because coincident defaults waste the protection (24, 25).

Basket / rainbow option A basket option pays on a weighted sum of several assets; a rainbow option pays on the best or the worst of them, as a worst-of put does. A basket is worth more as correlation rises and a worst-of put more as it falls, which is what pays the high coupon on a worst-of note (34, 35).

Bermudan option An option exercisable on a fixed set of dates. The standard embedded option in callable bonds and cancellable swaps (31, 33).

Beta A portfolio's sensitivity to an index, and the hedge ratio for hedging it with index futures: the number of contracts is beta times the portfolio's value divided by the value of one futures contract (3).

Binomial tree A discrete lattice for the underlying, and the most direct route to risk-neutral valuation (13).

Black's approximation Valuing an American call on a dividend-paying stock as the greater of two European calls: one expiring at the option's expiration on the dividend-adjusted price, and one expiring just before the ex-dividend date on the unadjusted price (15).

Black's model The Black–Scholes–Merton model applied to a forward rather than a spot; the market standard for caps, swaptions, and futures options (16, 29).

Black–Scholes–Merton differential equation The partial differential equation satisfied by every derivative whose value depends only on the stock price and time. The stock's expected return does not appear in it; the instrument enters only through its boundary condition (15, 20).

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Black–Scholes–Merton formula The closed-form value of a European call or put on a lognormal stock, $c = S_0N(d_1) - Ke^{-rT}N(d_2)$. In practice it is used mostly in reverse, to convert a market price into an implied volatility (15).

Bootstrap (statistics) Estimating the sampling error of a statistic by recomputing it on many samples drawn with replacement from the data itself. For a far quantile it is rough, because no resample can contain a loss larger than the largest observed (38).

Bootstrapping Building a curve by solving instrument by instrument from the short end outward (4, 7).

Borrow fee The fee a short seller pays to borrow the asset: a few basis points for a liquid stock, tens of percent for one that is hard to borrow. It does not by itself widen the forward-pricing relation, because holders can defend it by selling; where they will not or cannot sell, it turns that relation and put–call parity into bands, and puts trade rich to calls (5, 10, 11).

Boundary condition The condition that, together with the Black–Scholes–Merton differential equation, determines a particular derivative's value: $\max(S - K, 0)$ at expiration for a call, $S - K$ for a forward. The equation is common to all of them; the instrument is the boundary condition (15).

Box spread A synthetic long position at one strike combined with a synthetic short at a higher one, paying the difference between the strikes at expiration whatever the underlying does. A zero-coupon bond assembled from four options, used to borrow and lend at the rate its price implies (11).

Bump-and-revalue Computing a sensitivity by moving one input a small amount and repricing; costs one valuation per input, which is what makes it impractical for a model with many (32).

Business time A clock that runs on expected variance rather than minutes (an intraday volatility profile plus the weights of scheduled events), so that a short-dated option can be priced with the Black–Scholes–Merton formula by replacing $\sigma^2 T$ with the variance remaining (23).

Butterfly spread Long one option at each of two outer strikes and short two at the strike midway between them, paying nothing outside the outer strikes and most at the middle one. Because its payoff is never negative, call prices must be convex in the strike; it is also how the curvature of the smile is traded. On a yield curve the name means the same shape across tenors: one sign at two tenors against the opposite sign at a tenor between them, flat under a parallel shift and so a position on the curve's curvature (11, 12, 19, 38, 39).

Buy-out price A fixed price at which an obligation to surrender renewable certificates can be discharged in cash instead. It caps the certificate's price, and an option on the certificate inherits the cap (36).

C

Calendar spread In futures, a position long one delivery month and short another; a bet on the shape of the curve, not its level. In options, a near-dated option sold against a longer-dated one at the same strike, which profits from the near option's faster time decay (5, 12).

Callable bond A bond the issuer may redeem early at a fixed price. The investor is short an option, which is why its spread over the curve overstates the credit compensation; see option-adjusted spread (31).

Cap A strip of interest rate call options, one per period, protecting a borrower against rates above a strike (29).

Capital conservation buffer A layer of common equity of 2.5% of risk-weighted assets above the minimums. A bank that draws on it is not in breach, but its dividends and bonuses are restricted until it is rebuilt (27).

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Capital valuation adjustment (KVA) The cost of the regulatory capital a trade consumes over its life: the equity a bank must hold against the position because the rules require it, not because it chooses to. On an uncollateralized trade it can exceed all the other adjustments combined (9).

Caplet One period of a cap (29).

Capped variance swap A variance swap whose realized volatility is capped, conventionally at 2.5 times the strike for single stocks. It is a variance swap less a call on variance, so unlike the uncapped contract it must be priced with a model (22).

Carry trade Borrowing in a low-rate currency to invest in a high-rate one. Hedged, it earns nothing, by covered interest parity; unhedged, it earns the interest differential and bears the currency risk, and whether that pays is an empirical question rather than an arbitrage (5, 16).

Cash-and-carry Borrowing cash, buying the asset, and carrying it to delivery against a short forward; the arbitrage that caps the forward price. The reverse cash-and-carry sells the asset and buys the forward, and needs either a stock loan or a holder willing to sell (5).

Cash-flow-at-risk The shortfall of a company's operating cash flow over a quarter or a year below its forecast that is exceeded with a stated probability: the variant of VaR a nonfinancial company uses, because its exposure is to cash flow, not to a book's market value (38).

Cash-or-nothing / asset-or-nothing Digital options that pay, if the underlying finishes past the strike, a fixed amount or the asset itself. Their values are the two terms of the Black–Scholes–Merton formula, and a vanilla call is an asset-or-nothing call less K cash-or-nothing calls (34).

CDS spread The running premium rate on a credit default swap: the rate at which the premium leg, paid until a credit event or maturity, has the same present value as the protection leg. CVA uses it as a price, not as an estimate of the probability of default (9, 24, 25).

Central counterparty (CCP) A clearing house that becomes buyer to every seller and seller to every buyer, replacing bilateral exposure with exposure to itself (2).

Certainty-equivalent value The sure amount of cash an employee would accept in place of an option grant. For an undiversified holder it is well below the grant's fair value, a real cost of paying in options that appears in no account (17).

Charm The rate at which an option's delta changes with the passage of time, $\partial^2 f / \partial S \, \partial t$. Negligible for a monthly option; the dominant, clock-driven hedging flow for a same-day option (18, 23).

Cheapest-to-deliver The bond a short futures position will actually deliver, given the conversion factors (6).

Clean price A bond's quoted price, which excludes accrued interest (6).

Clearing member A well-capitalized firm with direct access to a CCP. Everyone else clears through a member as a client, so if the member fails, what a client recovers turns on how its positions and collateral were held; see segregation and porting (2).

Cliquet A strip of forward-starting options. The purest instrument for trading forward volatility, and a capped one is short volatility (34).

Close-out netting The contractual right to terminate all trades with a defaulted counterparty and settle a single net amount. The whole capital framework assumes it is enforceable (1, 7, 27).

CMS spread A coupon paying the difference between two swap rates of different tenors, the building block of steepener notes; its value turns on the correlation between the two rates (30, 31, 32).

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Coherent risk measure A risk measure that is monotonic, positively homogeneous, translation-invariant and subadditive. Expected shortfall is coherent; VaR is not, because it can fail subadditivity (38).

Collar A put bought and a call sold against a holding, the call's premium paying for the put; it fixes a band of outcomes. A zero-cost collar on a currency is a range forward (12, 16).

Collateral Assets posted to secure an exposure. What converts credit risk into liquidity risk (8).

Collateral valuation adjustment (ColVA) The value of the ways a real CSA departs from daily cash collateral that exactly tracks the mark-to-market: thresholds and minimum transfer amounts, non-cash collateral, the option to choose the collateral currency, and the margin period of risk (9).

Collateralized debt obligation (CDO) A tranched claim on a pool of bonds or loans. Named for its collateral: a CLO holds leveraged loans, a CDO-squared holds tranches of other deals (24, 26).

Collateralized loan obligation (CLO) A CDO whose collateral is leveraged corporate loans, actively managed within indenture constraints and protected by coverage tests that divert cash from equity to the senior notes (26).

Common equity tier 1 The highest-quality layer of bank capital: ordinary shares and retained earnings. DVA gains and losses are filtered out of it, and with the perpetual instruments that rank just behind it, it forms the tier 1 capital of the leverage ratio (27).

Complete market A market in which every payoff can be replicated by trading the available assets. An arbitrage-free market is complete exactly when its equivalent martingale measure is unique (the second fundamental theorem), and jumps and stochastic volatility make real markets incomplete (13, 28).

Composite option (compo) An option on a foreign asset whose foreign-currency payoff is converted at the exchange rate prevailing at expiration, so the holder bears both the asset and the currency risk: in effect an option on the asset measured in domestic currency, with volatility $\sqrt{\sigma_S^2 + \sigma_X^2 + 2\rho\sigma_S\sigma_X}$. Contrast the quanto, which converts at a fixed rate (16).

Compound option An option on an option, such as the right to pay a fixed amount at one date for a call expiring at a later one. It has a closed form under Black–Scholes, and its value depends on volatility twice over, through both the outer and the inner option (34).

Conditional value at risk (CVaR) Another name for expected shortfall, used in the optimization literature and as the training objective of deep hedging (38, TN39).

Constant-maturity swap (CMS) A swap one of whose legs pays a long swap rate (the ten-year rate, say), observed afresh at each reset and paid on a single date, rather than across the annuity the rate belongs to. That mismatch is what the convexity adjustment prices (28, 30, 33).

Contango A forward curve sloping upward (3, 5).

Continuity correction Pricing a barrier monitored m times as a continuously monitored one with the barrier shifted away from spot by the factor $e^{\pm 0.5826\,\sigma\sqrt{T/m}}$, the Broadie–Glasserman–Kou result (20, 34).

Continuous compounding The limit of compounding m times a year as m grows without bound, under which an amount A grows to Ae^{Rn} in n years. A mathematical convenience used throughout the option-pricing chapters, not a market convention (4).

Control variate Estimating a quantity together with a related one whose true value is known, by the same method, and correcting the first estimate by the error measured on the second. It works for trees and for Monte Carlo alike (20, 34).

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Convenience yield The benefit of holding a physical commodity rather than a claim on it; what makes a commodity forward curve invert (5, 36).

Conversion factor The exchange-published multiplier making bonds of different coupons roughly interchangeable for futures delivery (6).

Convertible bond A bond the holder may exchange for a fixed number of the issuer's shares, the conversion ratio. A corporate bond plus a call on the issuer's own equity, so the dilution analysis of warrants applies to the call (10, 25).

Convexity Curvature of value in the underlying variable. A source of both hedging error and systematic price adjustment (4, 30).

Convexity adjustment The correction applied when a rate is paid at a date, or settled in a way, other than the one under which it is a martingale: the gap daily margining opens between a futures rate and a forward rate, or a swap rate paid on a single date as in a CMS (6, 30).

Convexity trade Buying interest rate futures and paying fixed on a swap over the same period, collecting the convexity adjustment between the futures rate and the forward rate. It is not riskless: it loses when realized rate volatility exceeds the volatility the adjustment implies (30).

Copula A function that imposes a dependence structure on given marginals: marginals from the market, dependence from an assumption. It is first met in wrong-way risk, built for default correlation as the one-factor Gaussian copula, and used to price tranches (9, 24, 26).

Cornish–Fisher expansion A correction of a normal quantile for skewness and excess kurtosis. It turns the known moments of a delta–gamma P&L into a VaR without simulation, and is reliable when the skew is moderate (38).

Correlation reserve Part of a book's model value held back against the chance that a correlation that cannot be hedged has been misestimated; the standard treatment for quanto positions (16, 30).

Correlation risk premium A persistent gap between the correlation priced into options and the correlation subsequently realized. It arises for the same reason as the volatility risk premium, but far less trading is available to close it (32).

Correlation swap A contract paying realized average pairwise correlation against a strike; unlike a dispersion trade it cannot be replicated from options (22).

Cost of carry The cost of holding an asset: financing, plus storage, minus income. For an investment asset the forward price is the spot price grown at the cost of carry; for a consumption asset the convenience yield is subtracted (5).

Countercyclical buffer A buffer of 0 to 2.5% of risk-weighted assets in common equity that each jurisdiction raises when credit grows fast and releases in a downturn. It does not apply to the leverage ratio (27).

Counterparty credit risk The risk that a derivatives counterparty defaults while owing you money. Distinct from lending risk because the exposure is itself uncertain (27).

Covenant-lite Of a loan: carrying no maintenance covenant, a financial ratio the borrower must keep meeting while the loan is outstanding. Now standard in leveraged loans, so the early warning a covenant breach used to give arrives later (26).

Coverage test A test written into a CLO's indenture that redirects cash automatically as the portfolio deteriorates. The overcollateralization test compares collateral par with the par of a tranche and everything senior to it; on a breach, the equity's interest pays down the senior notes (26).

Covered bond A bank's bond secured on a pool of mortgages that stays on its balance sheet. The holder has dual recourse, to the bank and then to the ring-fenced pool, and no risk is

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transferred (26).

Covered interest parity The relation that sets a forward exchange rate equal to the spot rate adjusted for the two currencies' interest rates, with no reference to any view of where the currency is heading. Violated persistently since 2008; the deviation is the cross-currency basis (5, 7).

Cox process A Poisson process whose intensity is itself random, also called a doubly stochastic Poisson process. The reduced-form model of default: survival is the expectation of $\exp\left(-\int \lambda\right)$ over intensity paths, and because default arrives as a surprise, short-dated spreads do not vanish (24).

Crack spread The refiner's margin: the value of refined products less the crude they are made from, quoted as a 3-2-1 combination of gasoline and heating oil against crude. A refiner is long it and hedges by selling it in futures (36).

Credit conversion factor The share of an off-balance-sheet item, such as an undrawn commitment or a guarantee, that the standardized approach treats as drawn before applying the risk weight (27).

Credit default swap (CDS) A contract paying on a credit event in exchange for a periodic premium (25).

Credit event The contractually defined trigger for a CDS payment. Not the same thing as default, and the gap is Chapter 25's recurring theme.

Credit-linked note A funded CDS: the investor pays principal upfront and loses it on a credit event, so the protection buyer has no counterparty exposure and the investor carries a second exposure to the note's issuer (25).

Credit support The loss a securitization's pool absorbs before a given bond takes any. It comes from subordination (the tranches beneath the bond), overcollateralization (assets in excess of liabilities) and excess spread, the pool interest left after bond interest and fees (26).

Credit support annex (CSA) The schedule to the ISDA master agreement under which the parties post collateral against their exposure to each other. It sets eligible collateral, thresholds, haircuts and the rate paid on collateral, and through that rate it fixes the curve on which the trade is discounted (1, 7, 8).

Credit triangle The relation $s = \lambda(1 - R)$ between spread, hazard rate, and recovery; exact for a continuously paid premium (24).

Credit valuation adjustment (CVA) The market value of counterparty credit risk on a derivatives portfolio (9).

Credit VaR A high quantile of a credit portfolio's one-year loss minus its expected loss, which is priced and provisioned rather than held as capital. In a large portfolio it is set almost entirely by default correlation (24, 39).

Critical stock price The stock price at each date below which immediate exercise of an American put is optimal; it rises toward the strike as expiration approaches. Traced through time it is the early-exercise boundary, the free boundary of the pricing equation (11, 13, 15).

Cross-currency basis The spread on a currency swap that covered interest parity says should be zero and is not; in practice it prices the relative scarcity of funding in the two currencies (7, 33).

Cross-currency swap A swap exchanging payments in two currencies. Unlike a single-currency swap, the notional principals are exchanged, at the start and again at maturity; the spread on its floating legs is the cross-currency basis (7, 33).

Cross-gamma The part of a position's response that comes from two variables moving together.

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For CVA, exposure depends on rates and currencies while the charge depends on credit, so hedging the credit sensitivity alone leaves the interaction exposed (9).

Cross hedge A hedge using futures on an asset other than the one being hedged. A one-for-one position is then generally not optimal, and the minimum-variance hedge ratio applies (3).

CVA capital charge Regulatory capital held against the risk that CVA itself moves as credit spreads change, which was where most counterparty losses in the 2008 crisis came from. Computed by the basic approach (BA-CVA) or the sensitivity-based standardized approach (SA-CVA), and part of KVA (9, 27).

D

Dark spread The coal-fired counterpart of the spark spread: the power price less the cost of the coal burned to produce it (36).

Debit valuation adjustment (DVA) The mirror image of CVA: the value to a dealer of its own option to default, which rises as its own credit deteriorates. Recognized in accounting, filtered out of regulatory capital (9, 27).

Default A counterparty's failure to meet its obligations on any of its contracts, or its bankruptcy or insolvency. It belongs to the counterparty, not to a trade, and reaches all its liabilities at once through cross-default, acceleration and bankruptcy; creditors differ only in what they recover. For derivatives, an event of default under the ISDA Master Agreement. A legal event, not a market price: a downgrade is not a default (9).

Default fund A pool to which every clearing member contributes, held to cover losses larger than a defaulter's margin can absorb. In the default waterfall the defaulter's own contribution is used first, then the CCP's own capital, then the surviving members' contributions (2).

Default point The asset value at which a firm defaults: the face value of its debt in Merton's model. Distance to default is measured from it (24).

Default risk charge The part of the trading-book capital charge for jump-to-default on traded bonds, credit derivatives and equities, which spread sensitivities cannot see. Offsets between different issuers are recognized only in part (39).

Degree day The number of degrees by which a day's average temperature falls below (heating) or rises above (cooling) 65°F, summed over a month or season. The index on which most weather derivatives settle (1, 36).

Deliverable basket The set of bonds a short Treasury futures position may deliver, defined by a maturity range on the delivery date, with conversion factors putting them on a comparable footing. The short's choice within it is what makes one bond cheapest to deliver (6).

Delta The rate of change of an option's value with the underlying; the hedge ratio (18).

Delta hedging Holding delta units of the underlying against an option so the position is locally insensitive to price. The strategy the Black–Scholes–Merton model assumes (15, 18).

Detachment point The loss level at which a tranche is exhausted (25, 26).

Difficulty adjustment The protocol rule that resets how hard a block is to mine, at intervals, so that blocks arrive on schedule however much computing power is running (37).

Digital option An option paying a fixed amount or nothing. Has no finite replicating portfolio, so it is quoted at the cost of an overhedge (12, 34).

Dilution The fraction of a company passing to option or warrant holders on exercise. Measured by share count, by the grant's value against market capitalization, or, in the accounts, by the incremental shares of the treasury stock method; the three differ (17).

Dirty price The amount actually paid for a bond: the clean price plus accrued interest. Also called the cash price (6).

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Discount curve The curve used to present-value a trade's cash flows. Determined by the collateral agreement rather than by the rate the trade references (7, 8, 28).

Dispersion trade Short index volatility against long single-stock volatility; a position in implied correlation (22).

Displaced tree A tree for a stock paying a known cash dividend, built on the stock price less the dividend's present value, which recombines, with that present value added back at every node before the dividend date (20).

Distance to default How many standard deviations of asset value separate a firm from its debt; the durable output of a structural model, and the input a commercial implementation maps to a default frequency (24).

Dividend equivalent Cash or extra units matching the dividends paid on the shares while restricted stock units are unvested. Without them, an RSU's grant-date value is the share price less the present value of the dividends forgone (17).

Duration The sensitivity of a bond's price to a parallel yield shift, to first order (4).

DV01 The change in a position's value for a one basis point move in rates; the same measure as PV01, named for the currency it is reported in (8, 29).

Dynamic hedging Holding a position and trading repeatedly to restore its hedge as the Greeks drift. It becomes impossible where gamma diverges (near a digital's strike at expiration, or at a barrier option's barrier), and a static hedge, options bought once and held, replaces it (18).

E

Economic capital The capital a bank's own model says it needs to survive its worst year at a chosen confidence, typically 99.95% or more for credit, and allocated to loans and businesses to price them (24).

Effective expected positive exposure In the internal model method for counterparty capital, the running maximum of expected exposure averaged over the first year, treated as a loan-equivalent amount (27).

Effective federal funds rate (EFFR) The rate at which US banks lend reserves to one another overnight, published by the New York Fed. Not the benchmark, because volume collapsed after 2008, but it is the policy rate, and the fed funds–SOFR basis between it and SOFR trades through swaps (4).

Effective notional In SA-CCR, a trade's notional scaled by its supervisory delta (its direction, or an option's delta), its supervisory duration (an annuity at a fixed 5%) and its maturity factor. Within a hedging set the three maturity buckets are aggregated with correlations of 0.7 between adjacent buckets and 0.3 between the outer two, so offsets across buckets are only partial; multiplied by the supervisory factor, the result is the add-on (27).

Elasticity (of an option) The percentage change in an option's value for a 1% change in the price of the underlying, $\Omega = S\Delta/f$: the option's leverage. Largest for out-of-the-money options and negative for puts; for equity viewed as a call on the firm it is $N(d_1)V/E$ (18, 24).

Elicitability The property of a risk measure that some scoring function is minimized, in expectation, by its true value, so that competing forecasts can be ranked by average score. VaR has it; expected shortfall does not on its own, but VaR and ES are elicitable jointly (38).

Elliptical distribution A joint distribution under which every portfolio's loss has the same shape, differing only in location and scale; jointly normal and jointly Student's t losses are the standard cases. On such losses VaR is subadditive at any confidence level above one half (38).

Emissions allowance A permit to emit one metric ton of carbon dioxide under a cap-and-trade scheme. Bankable, storable at no cost and impossible to borrow, so its forward price is capped

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at spot carried at the financing rate but can sit below it, and its price is the marginal cost of abatement (1, 36).

Equilibrium model / no-arbitrage model An equilibrium short-rate model produces today's term structure as an output of its assumed process; a no-arbitrage model makes the drift time-dependent so that today's curve is an input, reproduced exactly, as a trading desk requires (31).

Equity / mezzanine / senior tranche The equity tranche takes a pool's first losses, the senior tranche suffers only after everything below it is exhausted, and the mezzanine tranches lie between. The equity tranche gains from correlation, and the senior tranche is short it, losing as correlation rises. A first-loss tranche is the equity tranche widened to detach at a chosen point (24, 25, 26).

Equivalent martingale measure For a chosen numeraire, a measure equivalent to the real-world one (agreeing on which events are impossible) under which every traded asset's price divided by the numeraire is a martingale. The risk-neutral measure is the one for the money market account (28).

Estimated initial value A structured note's fair value net of fees, which some jurisdictions require the issuer to print on the term sheet. It turns an argument about fairness into a number a buyer can compare across issuers (35).

Euler scheme Stepping a stochastic differential equation by its increments rather than by its solution. Exact only when drift and volatility are constant, as for arithmetic Brownian motion; available for any process; and it can drive a positive process below zero (20).

European Market Infrastructure Regulation (EMIR) The EU regulation that mandates central clearing of standardized OTC derivatives, margin on uncleared trades and trade reporting. Its 2024 revision requires EU firms above the clearing threshold in euro and zloty interest rate derivatives and euro short-term interest rate derivatives to hold an active account at an EU CCP and clear a minimum number of trades through it (2).

Event contract A contract paying a fixed amount if a stated event occurs and nothing otherwise: an inflation release, a central-bank decision, an election. A digital on an outcome no option spans, so it is priced as a probability rather than by replication (34).

Event of default Under the ISDA Master Agreement, a failure to pay or deliver not cured within the grace period, bankruptcy or insolvency, or an agreed cross-default, among others. It entitles the other party to terminate and close out every trade under the agreement (7, 9).

Excess kurtosis The fourth standardized moment less three, $\mathbb{E}[(X-\mu)^4]/\sigma^4 - 3$: zero for a normal distribution and positive when the tails are heavier. With skewness it is one of the two corrections in the Cornish–Fisher expansion (38).

Ex-dividend date The first day on which a buyer of a stock is no longer entitled to the coming dividend. The price falls by about the dividend, which can make exercising an American call the day before rational (10, 11).

Exchange option An option to give up one asset and receive another, valued by Margrabe's formula with the asset given up as numeraire. No interest rate enters, and the correlation between the two assets is what the value turns on (28, 34, 36).

Exercise Taking up an option's right. The decision, not the contract (10, 11).

Exotic option Any option that is not a standard call or put. The useful test is not complexity but whether the payoff can be replicated by traded instruments (34).

Expected default frequency (EDF) A default probability read from an empirical table indexed by distance to default, rather than from the normal tail the structural model implies (24).

Expected exposure (EE) / expected positive exposure (EPE) Expected exposure is the av-

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erage exposure across simulated scenarios at each future date, the average of $\max(V(t), 0)$; expected positive exposure is its time average, a single number: over the life of the trade as a summary of the profile, and over the first year in the capital rules, whose effective version averages the running maximum. CVA is priced on expected exposure (9, 27).

Expected shortfall (ES) The average loss conditional on exceeding the VaR level. Coherent where VaR is not (38).

Explicit / implicit scheme The two basic finite difference schemes. The explicit scheme reads each node directly from the next time step and is a trinomial tree in disguise; the implicit scheme solves a tridiagonal system at each step. Crank–Nicolson averages the two and converges faster (20).

Exponentially weighted moving average (EWMA) A volatility estimate that weights past squared returns geometrically, $\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1-\lambda)u_{n-1}^2$, with $\lambda = 0.94$ the RiskMetrics convention. It is GARCH(1,1) with no long-run level to revert to, so its forecast is flat at every horizon (38).

Exposure What a party would lose if its counterparty failed now, before recovery: the positive part of the portfolio's value, net of collateral. Not the notional, and not the value, which can be negative (9, 27).

Exposure at default (EAD) The regulatory measure of what is at risk when a counterparty fails (27).

Extreme value theory The theory of a distribution's extremes. Its central result, that losses beyond a high threshold follow a generalized Pareto distribution for almost any underlying law, lets a tail be fitted from a sample's worst observations (peaks over threshold) and extrapolated to quantiles the sample cannot reach. The fit assumes the tail's shape continues beyond the data, and the threshold is the analyst's choice (38).

F

Fair variance swap rate The strike at which a variance swap is worth zero today, the risk-neutral expectation of realized variance. Realized variance is replicated by a static portfolio of out-of-the-money options weighted by $1/K^2$ together with a dynamic position in the underlying, and the fair rate is the value of that strip grown at the risk-free rate; the construction behind the VIX (19, 22).

Feller condition The condition $2ab > \sigma^2$ under which the Cox–Ingersoll–Ross short rate stays strictly positive; when it fails, zero is reached and acts as a reflecting boundary (31).

Filtered historical simulation Historical simulation that divides each past return by that day's GARCH volatility and draws the standardized shocks at random through the GARCH recursion to build scenarios, due to Barone-Adesi, Giannopoulos and Vosper. It goes beyond volatility-scaled historical simulation by modeling how volatility evolves over the horizon (38).

Finite difference method Solving a derivative's pricing equation on a grid of prices and times, replacing its derivatives by differences and stepping back from expiration. The natural method for early exercise and for revaluing a whole book across a grid (20).

Flat price The futures price itself, as distinct from the basis. A hedge removes the flat-price risk and leaves the basis risk, and the two should be reported separately (36).

Flat / spot volatility A flat volatility is the single volatility which, applied to every caplet in a cap, reproduces the cap's price (a quotation device); a spot volatility, also called a caplet or forward volatility, is the volatility of one caplet. Spot volatilities are bootstrapped from flat ones, as zero rates are from par rates (29).

FLEX option An exchange-listed option whose strike, expiration and exercise style the parties

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choose, combining over-the-counter customization with a clearing house (10).

Flexi cap A cap on which only a limited number of caplets pay, the first that finish in the money. Path-dependent through the count, so it needs a term structure model (32).

Floor A strip of interest rate put options, protecting a lender against rates below a strike (29).

Forecast curve The curve from which forward rates are projected, distinct since 2008 from the curve used to discount them (7, 32).

Forward contract An agreement to trade an asset at a set price on a future date. Settled once, at maturity (1, 5).

Forward curve The settlement prices of successive delivery months plotted against those months, showing what the market will pay for the asset at every date it trades. Used by firms that never trade a contract. Its shape reflects the cost of carry and, for consumption commodities, a convenience yield (2, 5).

Forward freight agreement A forward fixing the cost of carrying a cargo on a given shipping route, settled against a published route index. Ship owners and charterers have the opposite exposures that let the market work (1, 36).

Forward points The difference between the forward and the spot exchange rate, in units of the last decimal place of the spot quote, which the client adds to spot to get the outright rate. Quoted in place of outright rates because they move far more slowly (5).

Forward price The delivery price that makes a forward contract worth zero at inception; it is not a forecast of the spot price, but the spot price carried to the delivery date at the cost of financing and holding the asset (1, 5).

Forward rate agreement (FRA) A forward contract on an interest rate (4, 6).

Forward-start option An option whose strike is set at a future date. Trivial under Black–Scholes and heavily model-dependent in practice, because it prices the forward smile (34).

Forward-starting variance swap A variance swap on a future window, replicated by a long position in a longer spot-starting variance swap and a short position in a shorter one, weighted by maturity. So the whole forward variance curve is spanned by today's vanilla surface (22).

Forward variance curve Today's price of the variance to be realized at each future time (forward, not spot, variance), read off the term structure of variance swap rates, the forward variance over an interval coming from two spot-starting swaps. What a stochastic volatility model has to reproduce, and the input the rough volatility models specify directly (21, 22).

Fractional Brownian motion A continuous Gaussian process whose variance over a horizon T is T^{2H}; increments are independent only at $H = 1/2$, where it is the Wiener process. Rougher than a Wiener path below one-half, and the model for volatility in rough volatility (14, 21).

Front-end protection In an index credit default swaption, the value of losses accumulated between trade date and expiration, owed to the protection buyer because the option does not cancel on default (28).

Full revaluation / partial simulation The two ways of computing Monte Carlo VaR: full revaluation reprices every position with its own model in each simulated scenario; partial simulation applies a delta–gamma approximation instead, which is faster and inherits that approximation's limits (38).

Fundamental Review of the Trading Book (FRTB) The Basel Committee's market risk capital standard, final in 2019: a prescriptive trading-book boundary, a sensitivities-based standardized approach, and internal models based on 97.5% expected shortfall with liquidity horizons, approved desk by desk (39).

Funding rate The periodic payment holding a perpetual futures contract in line with spot (37).

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Funding valuation adjustment (FVA) The cost of funding an uncollateralized derivatives position (9).

Futures contract An exchange-traded forward, standardized and settled daily (2).

Futures price The price at which a futures contract trades. It equals the forward price when rates are deterministic, and differs by a covariance between the asset and the financing rate when they are not (2, 5).

Futures rate The interest rate implied by an interest rate futures price, 100 less the quotation. Daily settlement makes it a martingale under the risk-neutral measure, so it exceeds the corresponding forward rate by the convexity adjustment (6, 30).

Futures-style margining Margining an option like a futures contract, with the premium posted and marked daily rather than paid up front. Black's model then loses its discount factor (16).

G

G2++ model A two-factor short-rate model in which the rate is the sum of two zero-mean, mean-reverting Gaussian factors and a deterministic shift fitted to today's curve. Rates at different maturities are then imperfectly correlated, so the curve can steepen and flatten (31).

Gamma The rate of change of delta with the underlying; the curvature a delta hedge cannot capture (18).

Gamma band The range of strikes over which an option's gamma exceeds half its at-the-money value. It narrows from several percent of spot a month before expiration to under half a percent in the final hour, which concentrates a same-day book's gamma at the current price (23).

General collateral / special A general collateral repo is one in which the cash lender accepts any security from an agreed class. A security in particular demand trades special, at a repo rate below general collateral and occasionally near zero, because the cash lender is giving up interest to borrow that security (4).

Generalized autoregressive conditional heteroskedastic (GARCH) model A volatility model; GARCH(1,1) updates the variance as $\sigma_n^2 = \omega + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$. The constant ω gives a long-run variance $V_L = \omega/(1 - \alpha - \beta)$ toward which the forecast reverts, which is what produces a volatility term structure; EWMA is the special case $\omega = 0$, $\alpha + \beta = 1$ (38).

Generalized Pareto distribution The law that the excesses over a high threshold of almost any loss distribution approach, with a scale and a shape parameter; the shape is positive for a power-law tail, zero for an exponential one, negative for a tail with an end point (38).

Generalized Wiener process A process $dx = a\,dt + b\,dW$ with constant drift a and a constant scale b on the Wiener noise; the $b\,dW$ term is the diffusion. Over a horizon T the change is normal with mean aT and variance b^2T (14).

Generator matrix The matrix Λ of rating-transition intensities with $P = e^{\Lambda}$, from which the transition matrix for any horizon is $e^{t\Lambda}$; the hazard rate extended from one exit to many (24).

Geometric average The nth root of the product of n fixings. The geometric average of lognormal prices is lognormal, so an option on it has a closed form, which makes it the control variate for the arithmetic Asian option (20, 34).

Geometric Brownian motion The process $dS = \mu S\,dt + \sigma S\,dW$, in which the proportional return has a constant expected rate and a constant volatility. It makes the stock price lognormal and is the model underneath the Black–Scholes–Merton formula (14, 15).

Girsanov's theorem The result that a change of measure changes a Wiener process's drift and leaves its volatility alone. Moving to a new numeraire, every process picks up a drift equal to the covariance between its logarithm and the log of the numeraire ratio, which generates the convexity, timing and quanto adjustments (28, 30).

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Global systemically important bank (G-SIB) A bank designated by size, interconnection and complexity, which holds an extra 1% to 3.5% of common equity against risk-weighted assets and half that again on its leverage requirement (27).

Gold lease rate What a holder earns by lending gold to a bullion bank, quoted as the financing rate less the forward premium on the metal. Its small size is what an essentially zero convenience yield looks like in a real market (5).

Greeks The sensitivities of an option's value to its inputs (18).

Gross basis / net basis For a deliverable bond, the gross basis is its quoted price less the futures price times its conversion factor, and the bond that minimizes it is the cheapest-to-deliver. The net basis subtracts carry, coupon income less repo financing, and measures the cost of the short's delivery options (6).

H

Haircut The discount applied to collateral's market value to allow for its own price risk (8).

Hang Seng China Enterprises Index (HSCEI) An index of mainland Chinese companies listed in Hong Kong. Its high volatility supported high coupons, and a large share of Korean retail autocallables referenced it (35).

Hash rate The computing power all miners together apply to producing blocks; a miner's share of it is its share of new coin (37).

Hazard rate The probability of defaulting in the next instant, given survival to it, per unit of time. A rate rather than a probability: the proportional rate at which the survival probability declines (24).

Heat rate The gas, in MMBtu, a power plant burns to produce one MWh. The spark spread is the power price less the heat rate times the gas price (36).

Heath–Jarrow–Morton (HJM) drift condition The no-arbitrage requirement that fixes the drift of every forward rate once its volatility functions are chosen: for each factor, the volatility times its integral out to the forward's maturity, summed over the factors. Short-rate models are special cases of the framework it defines (32).

Hedge accounting Accounting that lets a firm which documents a hedging relationship in advance and shows it to be effective defer the derivative's gains and losses (for a cash flow hedge of a forecast transaction, in other comprehensive income) until the hedged item affects profit, so the two sides meet in the same period (3).

Hedge ratio The size of the futures position relative to the exposure. The minimum-variance hedge ratio, $h^* = \rho\,\sigma_S/\sigma_F$, minimizes the variance of the hedged position; it is not one in general, and least of all in a cross hedge (3).

Hedging set In SA-CCR, a group of trades within an asset class whose effective notionals may offset: for interest rates, one per currency, divided into maturity buckets of under one year, one to five years and over five years (27).

Historical / hypothetical scenario The two sources of stress scenarios: the moves of an actual episode replayed on today's positions, or moves written by the firm for the risks it holds, which should change correlations and liquidity as well as prices (38).

Historical simulation Estimating VaR or expected shortfall by applying each of the last N days' market moves (500 is standard) to today's portfolio and reading the measure off the resulting profits and losses. It assumes no distribution, so fat tails and joint behavior come for free, but it cannot produce a loss larger than any in its window and reports a small number after a calm period (38).

Hurst parameter The exponent H that sets how the variance of fractional Brownian motion

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grows with the horizon, T^{2H}; one-half for the Wiener process, and about 0.1 for realized volatility across equity indices and liquid futures markets (14, 21).

Hypothetical / risk-theoretical P&L A desk's daily profit on yesterday's positions, computed by full front-office revaluation (hypothetical) and using only the risk factors in the risk model (risk-theoretical). The P&L attribution test compares the two (39).

I

Implied correlation The single correlation that, with the constituents' implied volatilities, reproduces an index's implied volatility; for a tranche, the correlation that reproduces its price. Demand for index protection keeps it above realized correlation, which is what a dispersion trade sells (19, 22, 25).

Implied repo rate The annualized return a cash-and-carry trade earns: buy the asset, finance it, and deliver it into the futures contract. Comparing it with the rate at which the position can actually be financed is how practitioners test a futures price (5, 6).

Implied volatility The volatility that makes the Black–Scholes–Merton formula reproduce a market price. It reflects the market's view of future volatility together with a risk premium, and across strikes and maturities it serves as a quoting convention rather than as a model of the stock (15, 19).

In-arrears swap A swap paying a rate at the start of the period it applies to; the classic convexity adjustment, and one the benchmark transition largely removed (33).

Incentive stock option (ISO) A US employee option carrying favorable tax treatment: no ordinary income at exercise, but the spread is an alternative minimum tax preference, and holding periods apply to keep the treatment. The AMT is what creates the exercise-and-hold trap: a tax bill on a paper gain in shares that cannot yet be sold (17).

Incremental VaR The change in a portfolio's VaR when one position is removed entirely. It answers whether to keep a whole position; unlike component VaR, the incremental figures do not sum to the total (38).

Index CDS A standardized CDS on a fixed portfolio of reference entities (25).

Index factor The fraction of an index CDS's original notional that remains after constituents have suffered credit events and dropped out: $124/125$ after one event in an index of 125 issuers. The coupon is paid on the reduced notional (25).

Index skew The difference between the level at which an index CDS trades and the intrinsic level computed from its constituents. It widens when the index is used heavily as a macro hedge (25).

Initial forward curve The instantaneous forward rates implied by today's discount factors, $f(0,t) = -\partial \ln P(0,t)/\partial t$; what a no-arbitrage short-rate model fits its time-dependent drift to (31).

Initial margin Collateral posted at the outset against potential future exposure, as distinct from variation margin against current exposure (2, 8).

Internal models approach (IMA) Computing trading-book capital from a bank's own expected shortfall model, permitted desk by desk once the desk passes backtesting and the P&L attribution test (39).

Internal ratings-based approach (IRB) Computing banking-book credit capital from the bank's own estimates of probability of default and, in the advanced version, loss given default, put through a Basel formula that is the worst-case default rate at 99.9% with a prescribed correlation and a maturity adjustment (39).

International Money Market (IMM) dates The third Wednesdays of March, June, Septem-

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ber and December, between which three-month SOFR futures settle. Swaps traded on these dates can be hedged with futures without date mismatches (6, 7).

Intrinsic value What an option would be worth if exercised now (10).

Inverse VIX product An exchange-traded note or fund delivering minus one times the daily return of a VIX futures index: a packaged short position in the front VIX futures that collects the roll carry and must buy futures into a rally to rebalance (22).

Investment asset / consumption asset An investment asset is held by enough investors purely for investment that they are indifferent between it and a claim on it: they will sell it, or lend it, when the forward is out of line, so arbitrage holds the forward price from both sides. A consumption asset is held to be used, so its holders will not part with it, only the upper bound survives, and the gap is filled by a convenience yield (5).

Investment grade / speculative grade Ratings of BBB (Baa) and above are investment grade; below that, speculative grade. A downgrade across the line forces sales by mandate-constrained holders and can trigger collateral calls (24).

Iron condor A put spread below spot and a call spread above it, sold together, which collects premium if the underlying stays within a range. Systematic selling of such structures leaves dealers long gamma in same-day options (12, 23).

ISDA master agreement The document under which OTC derivatives between two parties form a single legal contract. The basis for close-out netting (1).

Itô integral The integral $\int b\,dW$ against a Wiener process, defined as the limit of sums with the integrand evaluated at the start of each interval, before the increment is revealed. That choice makes it a martingale (14).

Itô process A process $dx = a(x,t)\,dt + b(x,t)\,dW$ whose drift and volatility may depend on its current value and on time; the form taken by the book's diffusion models, which the jump and rough volatility models of Chapter 21 go beyond (14).

Itô's lemma The chain rule for functions of a stochastic process (14).

J

Jamshidian's decomposition Pricing a European swaption in a one-factor short-rate model as a portfolio of options on zero-coupon bonds, each struck at the bond's value at the rate that makes the swap worth par. An identity, not an approximation, but it fails with a second factor or a Bermudan schedule (31).

K

KMV The standard commercial implementation of Merton's structural model, built by the firm Kealhofer, McQuown and Vasicek founded and now part of Moody's Analytics. It sets the default point at short-term debt plus half of long-term debt, and maps distance to default to an expected default frequency through an empirical table rather than the normal tail (24).

Knock-in / knock-out A barrier option created, or extinguished, on touching the barrier. Related by in-out parity (34).

L

Large-pool loss distribution The distribution of a large homogeneous portfolio's loss rate under the one-factor Gaussian copula. It underlies the Basel capital charge and shows why a senior tranche is a statement about correlation (26).

Law of iterated expectations Averaging a later conditional expectation over what is known earlier gives the earlier one: $E[E[X \mid \mathcal{F}_s] \mid \mathcal{F}_t] = E[X \mid \mathcal{F}_t]$ for $t \le s$. Also called the tower property (28).

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Legging risk The risk that the market moves between the fills of a multi-leg option strategy executed one leg at a time, which can exceed the value of the spread itself. Exchanges run complex order books that match the legs as a single package (10, 12).

Leptokurtosis More probability mass in the center and in the tails than a normal distribution carries. The equity smile is what it looks like when measured from option prices rather than assumed (19).

Leverage A position's notional divided by the capital posted against it; for one unit of a futures contract, the futures price divided by the margin on it, or one over the margin rate (1, 37).

Leverage function In a local-stochastic volatility model, the function of spot and time that multiplies the stochastic variance so that the vanilla surface is reproduced exactly; the square root of local variance divided by the conditional expectation of the stochastic variance (21).

Leverage ratio Tier 1 capital divided by total exposure (assets, derivative exposure, securities financing and off-balance-sheet items), with no risk weighting. A deliberately risk-insensitive constraint, so every trade consumes capital whatever its quality (27).

Liability-driven investment (LDI) A mandate that hedges a pension scheme's liabilities rather than tracking an asset benchmark, typically with leveraged rate exposure. The strategy at the center of the 2022 UK gilt episode (8).

LIBOR The interbank offered rate that anchored derivatives markets until its replacement by risk-free rates (4).

Likelihood-ratio method Estimating a sensitivity by differentiating the probability density rather than the payoff; handles a discontinuous payoff, at the cost of higher variance (20, 32).

Liquidation preference Preferred shareholders' right to be paid back first when a company is sold or wound up. Because employees' common stock sits behind it, a private company's 409A price is typically well below the price of its latest preferred round (17).

Liquidity horizon The time a regulator assumes it takes to exit or hedge a risk factor, from 10 to 120 days by class. Expected shortfall is scaled across the horizons, so a desk's capital depends on how its factors are classified (38, 39).

Local volatility A model in which volatility is a deterministic function of spot and time; fits any surface exactly and gets the forward smile wrong (21).

Lockout period The initial period of an autocallable during which the note cannot be called (35).

Log contract The payoff $-2\log(S_T/F)$; delta-hedging it earns realized variance on every path, and it is replicated by a strip of options weighted $1/K^2$, which is why variance swaps are replicable (22).

Lognormal distribution The distribution of a variable whose logarithm is normal. Geometric Brownian motion makes the terminal stock price lognormal, and a binomial tree of small independent multiplicative moves converges to it (13, 14).

Long / short gamma A book long gamma gains from a move in either direction and hedges by selling rallies and buying dips, damping the move; a book short gamma, having sold options, hedges by trading with the move and amplifies it (18, 23).

Long-term equity anticipation securities (LEAPS) Listed options with expirations up to about three years on single stocks and five on the S&P 500 (10).

Lookback option An option on the maximum or minimum of the path. Taught more than it is traded (34).

Loss given default The fraction of a claim a creditor loses when the borrower defaults, $1 - R$. Conventionally 60% for senior unsecured debt, and not separable from the default probability

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in what a CDS spread observes (9, 24).

M

Mahalanobis distance The distance of a joint move $\mathbf{r}$ from the center of a distribution with covariance Σ, $\sqrt{\mathbf{r}^\top \Sigma^{-1} \mathbf{r}}$, counted in standard deviations along whatever direction the move takes. Under normality the density falls as it grows, so the most plausible scenario producing a given loss is the one that minimizes it (38).

Maintenance margin The balance below which a futures account triggers a margin call, set below the initial margin; the trader must then restore the account to the initial level (2).

Margin call A demand for additional collateral: on a futures account, when the balance falls below the maintenance level; under a CSA, when one party's valuation of the portfolio says the exposure has grown (2, 8).

Margin period of risk The assumed time between a counterparty's last margin payment and the close-out of its positions, conventionally five days for cleared trades and ten for bilateral ones (8, 27).

Margin valuation adjustment (MVA) The cost of funding initial margin over the life of a trade. Initial margin is posted gross, segregated and cannot be reused, so it must be funded from inception to maturity (8, 9).

Marginal / component VaR Marginal VaR is the derivative of VaR with respect to a position, $\partial \mathrm{VaR}/\partial w_i$; component VaR is the position times its marginal VaR. Because VaR is homogeneous of degree one in the positions, the components sum exactly to the total (an Euler allocation), though that does not make VaR subadditive (38).

Margrabe formula The closed form for an option to exchange one asset for another. Contains no interest rate, because the second asset is the numeraire (28, 34).

Market agreed coupon (MAC) A standardized swap whose fixed coupon is a round number in 25 basis point increments, the difference from the par rate being paid upfront. One product among many, not the general form of a swap: it buys fungibility and therefore cheaper compression and clearing, at the cost of cash at inception (7).

Market price of risk The excess return per unit of volatility, $(\mu - r)/\sigma$, that no-arbitrage forces to be the same for every asset driven by one source of uncertainty. It is unique only when the traded assets are at least as numerous as the sources of uncertainty they must hedge (28).

Marking to market Revaluing a position at current prices, and for futures, settling the change in cash daily (2).

Markov process A process in which only the present value matters for predicting the future; the path that led there carries no information. Assuming stock prices are Markov is a statement of weak-form market efficiency (14).

Martingale A process with no drift under a given measure. Choosing the measure that makes the right quantity a martingale is the technique of Chapter 28.

Maturity adjustment The factor in the IRB formula that raises the capital on a longer loan for the downgrade risk it carries before it can default. It equals one at a one-year maturity (39).

Maturity factor In SA-CCR, the scaling of a trade's effective notional to the horizon over which it is exposed: the square root of its remaining maturity, capped at one year, when unmargined, and a function of the margin period of risk when margined (27).

Maximum likelihood Estimating a model's parameters as those under which the observed data are most probable, by maximizing the logarithm of their joint density. It fits the generalized Pareto tail and the GARCH parameters in Chapter 38 (38).

Mean reversion The tendency of a rate to be pulled toward a long-run level, which distinguishes

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rates from equities. Most short-rate models have it; Ho–Lee does not, which is what rules it out in practice (31).

Minimum transfer amount The smallest collateral call a CSA permits; below it no transfer is made. With the threshold, it is what leaves an exposure uncollateralized between calls (8, 27).

Mixing parameter The scale on the volatility of volatility in a local-stochastic model, between zero (pure local volatility) and one; every setting fits the vanillas, so it measures model risk rather than fit (21).

Model risk The risk that the model mapping market factors to a position's value is wrong. It is not a subset of market risk, because an error in the model moves the measure and the position together (21, 39).

Modified duration Duration divided by $1 + y/m$: the sensitivity of a bond's price to its yield when the yield is quoted with compounding frequency m rather than continuously (4, 29).

Moneyness How far in or out of the money an option is; several conventions exist and they are not interchangeable (19).

Monte Carlo VaR VaR and expected shortfall read off losses simulated from the joint distribution of the risk factors. The only reliable route for strongly nonlinear or path-dependent books, at the cost of sampling error (38).

Multi-curve framework Discounting on one curve and projecting forwards on another, as practice has required since 2008 (7).

Multiple For a catastrophe bond, the ratio of its spread to its modeled expected loss. It falls as the expected loss rises, so remote risks carry proportionally more compensation (36).

Multiplier In SA-CCR, a factor between 0.05 and one that reduces potential future exposure when collateral exceeds the current exposure or the netting set is out of the money; it equals one otherwise (27).

N

Naked option An option written without a position in the underlying that covers it, such as a call sold without owning the shares. A naked call writer's exposure is unlimited and a naked put writer's is bounded by the strike; both must be collateralized with margin (10, 18).

Natural measure / required measure A variable's natural measure is the one under which its forward value is an unbiased expectation, set by the numeraire in its definition; the required measure is the one the contract's payment date and currency call for. Every convexity, timing and quanto adjustment is the covariance that moving from the first to the second introduces (30).

Negative convexity The property of a security whose price gains less in a rally than it loses in a sell-off, as a mortgage pass-through's does because borrowers prepay when rates fall. Hedging it means trading in the direction the market is already moving (6, 26).

Net open interest In a CDS auction, the amount of bonds still wanted for purchase or offered for sale once offsetting physical settlement requests are matched. The auction's second stage clears it, and the clearing price is the final recovery rate (25).

Netting opinion A legal opinion, commissioned by the industry for each jurisdiction, that close-out netting will be upheld in a counterparty's insolvency there. Where netting is not enforceable, exposure and capital revert to the gross figure (27).

Netting set The trades with one counterparty that a single enforceable netting agreement covers. The unit at which exposure is measured (8, 27).

No free lunch with vanishing risk (NFLVR) The strengthening of "no arbitrage" that is equivalent, in continuous time, to the existence of an equivalent local martingale measure (28).

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Non-modellable risk factor (NMRF) A risk factor that fails the risk factor eligibility test for lack of real price observations, and is capitalized by a stress scenario added to the modeled charge rather than diversified with it (39).

Non-qualified option (NSO) The ordinary US employee option, taxed as ordinary income on the spread at exercise and withheld through payroll. No alternative minimum tax complication and no holding period, so the choice against an incentive stock option is a genuine trade-off rather than one dominating the other (17).

Normal (Bachelier) volatility The standard deviation of a rate's change measured in basis points, as in Bachelier's model, rather than of its percentage change. The standard quote for rate options since policy rates went negative; roughly the lognormal volatility times the rate level (4, 19, 29).

Notional The reference amount on which payments are computed. Not an amount anyone owes or can lose, but not meaningless either: it scales operational load and is the base for several regulatory measures (1).

Novation The replacement of a contract between two parties by two contracts, each with a central counterparty. It is what allows a futures position to be closed out against anyone and makes contracts fungible whatever the identity of the seller (2). Bilaterally, a third party stepping into one side of an existing trade, which keeps the trade but changes the counterparty and so requires the remaining party's consent (33).

Numeraire The asset in whose units prices are quoted. Changing it is the central technique of Chapter 28.

O

One-factor Gaussian copula The standard copula for default correlation: each borrower's latent variable is $\sqrt{\rho}\,M + \sqrt{1-\rho}\,Z_i$, with M a common factor, and the borrower defaults when it falls below the threshold set by its own default probability. Conditional on M the defaults are independent, which is what makes the large-pool loss distribution tractable (24, 26).

One-touch / no-touch option A one-touch pays a fixed amount if the underlying trades at a level at any time before expiration; a no-touch pays it if the level is never reached. Standard dealer products in foreign exchange (34).

Open interest The number of contracts outstanding. Measures what is held, where volume measures turnover (2).

Option-adjusted spread (OAS) The spread that reconciles a callable bond's model price with its market price (29, 31).

Option chain The grid of a listed underlying's options, strikes down and expirations across, showing bid, ask, volume and open interest for each call and put (10).

Ornstein–Uhlenbeck process A Gaussian process pulled back toward a fixed level at a constant rate, the dynamics Vasicek's model gives the short rate. In the Hull–White form of HJM the whole forward curve is a function of one such process (31, 32).

Output floor The rule that a bank's total risk-weighted assets from internal models may not fall below 72.5% of what the standardized approaches give, across all risk types. A bank on the floor gains nothing from a better model (39).

Overhedge Selling a discontinuous payoff and hedging with a portfolio whose payoff is at least as large everywhere, paying a known premium in place of an unhedgeable jump. That premium sets the floor under the dealer's quote, which in practice is wider (34).

Overnight indexed swap (OIS) A swap exchanging a fixed rate for the compounded overnight rate. The instrument from which the discount curve is built (4, 7).

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P

Par rate The fixed rate making a swap worth zero at inception (4, 7).

Par yield The coupon rate at which a bond is priced at its face value (4).

Participation rate The share of an underlying's rise that a structured note pays, which may exceed 100%. A residual rather than a promise: it is set by how many calls the money left after buying the zero-coupon bond will buy, and so mostly by the issuer's funding cost (35).

Pass-through security A mortgage-backed security that passes the pool's monthly interest and principal, prepayments included, straight to investors in proportion to their holdings. An agency pass-through carries little credit risk; its risk is prepayment, which is an option problem (26).

Path dependence A payoff that depends on the route, not only the destination. Where the model risk in exotics lives (34).

Pathwise differentiation Estimating a sensitivity by differentiating the payoff along each simulated path, with the draws held fixed, and averaging. It uses the paths that produced the price and leaves no bump size to choose, though its variance is no lower than a bump on common random numbers; it requires a differentiable payoff, so it works for a call and fails for a digital (20, 32).

Payment for order flow Payment by a wholesale market maker to a retail broker for routing customer orders to it, which has funded zero-commission trading in the US since 2019. The cost moves from an explicit commission into the execution price (10).

Payoff What a contract delivers at expiration or settlement, before any cost of acquiring it: $\max(S_T - K, 0)$ for a call, a note's redemption, the amount a swap pays. Distinct from profit (1, 10).

Peaks over threshold Estimating an extreme quantile or expected shortfall by fitting the generalized Pareto distribution to the losses above a high threshold and extrapolating the fitted tail beyond the data (38).

Perpetual futures A futures contract with no delivery date, held near spot by a funding-rate mechanism (37).

Physical measure The real-world probabilities that generate observed data, as distinct from the risk-neutral measure under which prices are expectations (21).

Pillar 1 / 2 / 3 The three parts of the Basel framework: the formula minimums on risk-weighted assets, the supervisor's bank-specific add-on for risks the formulas miss, and disclosure (27).

Pin risk The uncertainty, for the writer of an option struck near the closing price at expiration, about whether it will be assigned. The writer learns the outcome only after the market has closed and is left with an unhedged position of unknown size and direction, which is why such positions are closed before expiration (10, 23).

Pinning The tendency of an underlying to be drawn toward a strike with large open interest as expiration approaches, because dealers long gamma there sell rallies and buy declines around it. With daily expirations it now operates every session (10, 18, 23).

Pip One unit in the last decimal place of an exchange-rate quote: 0.01 yen in a dollar–yen quote (5).

P&L attribution test The test of whether a desk's risk model captures its P&L: hypothetical and risk-theoretical P&L are compared over 250 days by Spearman rank correlation and the Kolmogorov–Smirnov statistic, placing the desk in a green, amber or red zone (39).

Plain vanilla swap The standard interest rate swap: fixed payments exchanged for floating payments on one notional principal in one currency, the notional itself never changing hands

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(7).

Poisson process A count of events that rises by one at random moments, at a given average rate. It drives the jumps in Merton's model and the arrival of default in reduced-form credit models (21, 24).

Portfolio insurance Replicating a protective put on an equity portfolio by selling index futures as the market falls and buying them back as it rises. Its mechanical selling into the decline of October 19, 1987 is the standard example of a dynamic hedge amplifying the move it hedges (16, 18).

Portfolio margining Setting margin on the net risk of a whole account, by stressing it across a grid of underlying prices and volatilities and charging for the worst outcome, rather than leg by leg. It substantially reduces requirements for hedged positions, and so permits larger ones (2, 10, 12).

Porting Transferring a failed clearing member's client positions, with their collateral, to a surviving member rather than closing them out. It preserves the client's hedge, and it needs margin computed gross, client by client, so that each client's collateral can move with it (2).

Potential future exposure (PFE) An estimate of how large an exposure could become over a horizon (9, 27).

Predictor–corrector An Euler step taken twice: once to reach a predicted point, then again with the drift averaged between the start and that point. The remedy for a state-dependent drift, at one extra drift evaluation per step (20, 32).

Premium-adjusted delta In foreign exchange, an option's delta less its premium expressed in units of the foreign currency, $\Delta - c/S_0$: the delta seen by a holder who pays the premium in that currency (16).

Prepayment risk The risk that mortgage borrowers refinance early, shortening a pass-through's life exactly when that hurts (26).

Price discovery The process by which trading assembles many participants' scattered information into a publicly visible price, a public good produced as a by-product of speculation (2).

Price limit A limit that halts trading in a futures contract if its price moves more than a set amount from the previous settlement. It does not remove the information driving the move, and a market locked at its limit is one in which no one can trade (2).

Prime broker A bank that finances and executes trades for hedge funds and similar clients, often through total return swaps. Archegos built positions that no one of its prime brokers could see in full (1, 27, 40).

Principal component analysis Decomposing the correlation matrix of curve moves into eigenvectors, the factors, and eigenvalues, their shares of variance. For interest rates the first three read as level, slope and curvature (32).

Principal-protected note A note that returns its full face at maturity plus a participation rate times any rise in the underlying: a zero-coupon bond plus a call, paid for with the interest the buyer gives up (12, 35).

Procyclicality The tendency of a requirement to move with the market cycle and so reinforce it. Margin rises when volatility rises, which is in a falling market, so every member must find cash at once and the natural source is selling into the decline (2, 8, 40).

Profit The payoff net of what the position cost. A profit diagram and a payoff diagram cross zero at different points: the breakeven and the strike respectively (1, 10, 12).

Prudent valuation Valuing a position at a price it could be realized at with 90% confidence

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rather than at fair value, and deducting the difference from equity. A rule in the EU and the UK (39).

Put–call parity The arbitrage relation linking a call, a put, the underlying, and a bond. Model-free (11).

Q

Quadratic variation The limit of the sum of a process's squared increments. For a Wiener process over $[0, T]$ it is exactly T, and Itô's lemma can be derived from it (14).

Qualified investor An investor meeting asset or subscription thresholds that permit the sale of complex products, such as China's snowballs, which are sold with minimum subscriptions of about RMB 1 million (35).

Qualifying CCP A central counterparty licensed and supervised under rules consistent with the international standards for clearing houses. Trades against one carry a 2% risk weight (27).

Quanto A payoff computed from an underlying denominated in one currency and paid in another at a fixed exchange rate agreed in advance, so the holder carries no currency exposure. Requires a correlation-driven adjustment to the underlying's drift (16, 30).

Quanto spread The difference between dollar- and euro-denominated CDS on a euro-area sovereign: the market's price for the currency's depreciation conditional on that sovereign's default (25).

Quasi-Monte Carlo Simulation with low-discrepancy sequences, such as Sobol or Halton, that fill the space more evenly than random numbers. Convergence can approach $1/M$ rather than $1/\sqrt{M}$, but the usual standard error no longer applies (20).

R

Radon–Nikodym derivative The random variable $\Lambda = dQ/dP$ that reweights one measure into an equivalent one, positive with $E^P[\Lambda] = 1$, so that $E^Q[X] = E^P[\Lambda X]$. Changing measure is multiplying by it before averaging, never a change in the possible outcomes (28).

Range accrual A coupon accruing only on days a reference rate is inside a band; a strip of digitals, and short volatility (33).

Range forward A zero-cost collar on a currency. An importer buys a call and sells a put, with strikes set so the premiums offset; an exporter does the reverse. Either locks in a band of exchange rates without paying a premium (16).

Ratchet cap A cap whose strike resets each period to the rate just observed plus a spread. Path-dependent in the forward-rate vector, and sensitive to the shape of instantaneous volatility that caplet prices leave undetermined (32).

Rebate A fixed amount paid to the holder of a knock-out option when it is extinguished, or at expiration if the barrier has been touched; valued separately from the option (34, 35).

Recovery rate The fraction of face value a defaulted claim realizes; in a CDS, set by auction (9, 24, 25).

Reference entity The borrower a CDS is written on. Not a party to the contract (24, 25).

Reference rate A published benchmark that contracts point to rather than negotiate individually. It is useful only if widely accepted and hard to manipulate; LIBOR satisfied the first condition and not the second (4).

Reinvestment period The period, typically about five years, during which a CLO reinvests the principal it receives rather than returning it to noteholders, within the indenture's eligibility and diversity constraints (26).

Replacement cost What it would cost to replace a contract on current terms if the counterparty

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defaulted: its value to the surviving party, net of collateral, when positive. Far smaller than a swap's notional, and the first component of exposure at default under SA-CCR (7, 27).

Repo A sale with an agreement to repurchase; economically a secured loan, and the mechanism behind much collateral funding (8).

Repudiation / moratorium Credit events for sovereign CDS: the government disowns its debt, or declares a suspension of payments on it. Contracts on emerging-market sovereigns also include obligation acceleration, a default on one obligation that lets its holders demand immediate repayment (25).

Residual risk add-on A flat charge in the standardized approach on payoffs its sensitivities do not describe: 1% of notional for exotic underlyings, 0.1% for other residual risks such as gap or correlation risk (39).

Restructuring The most contested credit event, because the entity survives and its obligations trade at very different prices (25).

Reverse convertible A note paying a high fixed coupon whose face is repaid in shares, taking the full fall, if the underlying ends below a barrier. A bond plus a written put; the coupon is the risk-free rate plus the put's premium, less the issuer's fee (12, 35).

Reverse knock-out A knock-out option whose barrier lies on the side where the option is in the money, such as an up-and-out call. It pays only in the band between strike and barrier, where the volatility surface constrains models least, and behaves like a digital at the barrier (21, 34).

Reverse stress testing Fixing a loss (one that would breach a limit or exhaust capital) and finding the scenario that produces it. For a linear book with normal factors the most probable such scenario moves each factor in proportion to its covariance with the portfolio (38, 40).

Rho The sensitivity of an option's value to the interest rate. On short-dated equity options it is of the same order as vega and still ignored, because rates barely move; it is first-order for long-dated and currency options (18).

Richardson extrapolation Combining values at n and $2n$ steps as $2V(2n) - V(n)$ to cancel an error proportional to $1/n$. Reliable on a tree only along one branch of its oscillation, once a control variate has left that branch decaying smoothly, and only when both step counts place the strike at the same offset between nodes (20).

Right-way risk Exposure that falls as the counterparty's credit deteriorates, the reverse of wrong-way risk; it reduces CVA (9).

Risk-adjusted return on capital (RAROC) A loan's or a business's return divided by the economic capital allocated to it, compared with a hurdle rate. It is how a portfolio's credit VaR enters the pricing of each loan (24).

Risk factor eligibility test The test of whether a risk factor may be modeled: at least 24 real price observations over twelve months with no 90-day period containing fewer than four, or at least 100 observations (39).

Risk-free rate (RFR) An overnight rate built from actual borrowing transactions, such as SOFR or €STR; the replacement for LIBOR. Contracts use it compounded in arrears over each interest period (4).

Risk measure A rule mapping a loss distribution to a single number, used as a limit, a capital requirement or a hedging objective. VaR and expected shortfall are the two in regulatory use (38, TN39).

Risk-neutral probability The probability under which every asset is expected to earn the risk-free rate. Not the real-world probability or anyone's forecast; it is what makes a derivative's value a discounted expectation (13).

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Risk-neutral valuation Pricing as a discounted expectation under a measure in which all assets drift at the risk-free rate. Not a claim about probabilities (13, 15).

Risk reversal An out-of-the-money call against the corresponding put. The FX market quotes it as the difference between their implied volatilities, the 25-delta call's less the 25-delta put's, which is how it quotes skew; as a position, it is the structure that trades skew (12, 16, 19).

Risk-weighted assets (RWA) Exposure multiplied by a supervisory risk weight set by counterparty type. Multiplied by the required capital ratio, it gives the capital a position consumes (27, 40).

Risky annuity The present value of one basis point paid until default or maturity, whichever comes first. Under the standard fixed-coupon CDS it carries the whole valuation, and it is the credit market's duration, shrinking as the credit worsens because default cuts the premium stream short (25).

Roll yield The gain or loss from rolling a futures position along the curve (5).

Rough volatility The finding that the path of an asset's volatility is far more jagged than a Wiener path, with a Hurst parameter near 0.1, and the models built on that finding, which reproduce the steep short-dated skew (14, 21).

S

SABR model A stochastic volatility model for a forward rate, $dF = \alpha F^{\beta}\, dW_1$ and $d\alpha = \nu\alpha\, dW_2$, fitted strike by strike to each swaption smile in the volatility cube: α sets the level, β the backbone, ρ the skew and ν the curvature. The market standard because of Hagan's closed-form approximation for the implied volatility it produces (29).

SA-CCR The standardized regulatory approach to counterparty exposure (27).

Samuelson maturity effect The tendency of a futures contract's volatility to rise as its expiration approaches, so that near contracts are more volatile than far ones. It follows from mean reversion in the spot price (36).

Scenario analysis Revaluing a whole book in full over a grid of moves in the underlying and in volatility and reading the worst cell, because the Greeks are local approximations that fail for large moves (18).

Second fundamental theorem of asset pricing An arbitrage-free market is complete exactly when its equivalent martingale measure is unique. In an incomplete market a model chooses a price rather than reading it from the market (28).

Segregation Keeping client collateral separate from a clearing member's own. Under individual segregation each client's collateral sits in its own account; under omnibus segregation clients are pooled, which is cheaper and offers less protection, since one client's shortfall can reach the others (2).

Segregation of duties The control that keeps the people who take positions apart from those who settle, reconcile and value them, so that no one checks their own work. Its absence, not any instrument, produced the Barings and Société Générale losses (40).

Self-financing portfolio A portfolio whose value changes only because the prices of its holdings change: rebalancing is allowed, but every purchase is paid for by a sale and no cash is added or withdrawn. The condition that makes a replication argument an argument about prices (28).

Sensitivities-based method The core of the standardized approach: prescribed risk weights and correlations applied to delta, vega and curvature sensitivities, computed under three correlation scenarios with the worst taken (39).

Sensitivity analysis Moving one risk factor at a time and reporting the change in value. Easy

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to run and explain, it misses the joint moves that make a crisis costly (38).

Shifted lognormal Black's model applied to a rate plus a constant displacement, which restores a usable quoting convention when rates are negative (29, 32).

Short hedge / long hedge A short hedge sells futures to protect an asset owned, or to be owned, against a fall in price; a long hedge buys futures to protect a planned purchase against a rise (3).

Short-rate model A model specifying the dynamics of the instantaneous rate, from which the whole curve follows (31).

SIMM ISDA's Standard Initial Margin Model, the industry method for computing initial margin on uncleared derivatives from weighted sensitivities. A common model as much to prevent disputes as to measure risk (8).

Skew Implied volatility falling as the strike rises, the shape of equity index and single-stock option prices since October 1987. Low-strike puts are expensive because the market assigns more probability to large falls than the lognormal allows; traded with the risk reversal (15, 19, 21).

Skewness A measure of a distribution's asymmetry, $\mathbb{E}[(X-\mu)^3]/\sigma^3$. For returns it is negative when the long tail lies on the loss side; for a distribution of losses the same tail makes it positive. An autocallable's return distribution is negatively skewed even when its underlying's is positively skewed (35, 38).

Smile The pattern of implied volatility across strikes, and evidence that the Black–Scholes assumption is wrong in a stable way (19).

Snowball The domestic Chinese form of the autocallable, path-dependent through its monthly knock-out and knock-in observations, and sold through asset-management products with a minimum subscription around RMB 1 million, so its buyers are qualified investors rather than depositors (35).

SOFR Index An index, published by the New York Fed, that accumulates daily compounded SOFR from a fixed start date, so the compounded rate between any two dates follows from two index values (4).

Sovereign credit The risk that a national government does not pay, chiefly on debt in a currency it does not issue, since it can always print its own. Default is a choice rather than an insolvency, there is no bankruptcy court, and local-currency debt can be inflated away, so structural models do not apply (24).

Spark spread The power price less the heat rate times the gas price: the margin of a gas-fired plant, which is therefore a strip of exchange options on it. Its value turns almost entirely on the correlation between power and gas (28, 36).

Special purpose acquisition company (SPAC) A listed shell company that raises cash from public investors to merge with a private operating company. Its units of a share and a fraction of a warrant made SPAC warrants the highest-volume recent example of dilutive warrants (17).

Spike model A model of power prices as ordinary lognormal days punctuated, with some probability, by spike days on which the price is multiplied by a lognormal jump. The price is a mixture of two lognormals, with a far heavier upper tail than a lognormal of the same mean and standard deviation (36).

Spot measure The measure whose numeraire is an investment rolled at each reset date into the bond maturing at the next one; an alternative to the terminal measure for simulating the LIBOR market model (32).

Spot-starting cap A cap whose first accrual period begins today. Under LIBOR conventions its first caplet, already fixed, was excluded; with backward-looking rates it is no longer trivial (29).

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Spring-loading Granting an option or other award just before good news is released, so that the grant price is set below where the news will take the stock. The successor to backdating: SAB 120, issued in 2021, requires the award's fair value to reflect the pending news, and Item 402(x), adopted in 2022, requires US companies to disclose grants made around such releases (17).

Stablecoin A token its issuer promises to redeem for one unit of a currency, usually a dollar, backed by a reserve of assets in that currency; the settlement asset of most crypto trading, and a claim on the issuer's credit (37).

Stack and roll Hedging a long-dated exposure by placing all of it in the nearest futures contract and rolling it into the next as each expires. Each roll is taken at whatever spread then prevails, which in contango costs money on the whole stack; the futures losses are cash today while the offsetting gains arrive over years; and a one-for-one stack is the wrong ratio, overhedging once discounting and the Samuelson maturity effect are counted (3, 6, 36).

Standardized approach (SA) The trading-book capital charge every bank computes from prescribed formulas: the sensitivities-based method, the default risk charge and the residual risk add-on (39).

State price The price today of a security that pays one unit if a particular state occurs and nothing otherwise, an Arrow–Debreu security. A market admits no arbitrage if and only if a set of positive state prices exists (13).

Static replication Hedging with a portfolio of vanillas set up once and traded again at most once (for a barrier, when it is touched), with no rebalancing in between. Where it is possible, model risk is small (34).

Sticky cap A cap whose strike resets to the previous *capped* rate plus a spread, so the strike follows the rate down but rises by at most the spread each period (32).

Sticky strike / sticky delta Competing assumptions about how the smile moves when spot moves. They imply different deltas for the same option (19).

Stochastic volatility A model in which volatility is itself random, producing a smile through the correlation between volatility and the underlying and through the volatility of volatility. Heston's version fits a surface badly at both ends: its skew is too flat at short maturities and decays faster than the market's at long ones (21).

Stock appreciation right (SAR) An award paying the spread between the share price and a strike, in cash, without the holder finding the strike. The same payoff as an option and none of the dilution, but cash settlement makes it a liability, remeasured to fair value every reporting date rather than fixed at grant (17).

Straddle A call and a put at the same strike. At the money it has almost no directional exposure, but only while the underlying stays near the strike; once it moves the position becomes directional, so a pure volatility position must be delta-hedged (12).

Straight-line / graded recognition Two ways to spread an equity award's grant-date value over the vesting period: straight-line treats the grant as one award expensed equally each year; graded treats each vesting tranche as a separate award, so the early years carry a share of every tranche (17).

Stress capital buffer A US capital requirement on top of the minimum, set by the losses a large bank projects in the supervisory stress test, including a global market shock to its trading book (39).

Stress test Revaluing a portfolio under a specified scenario of moves in prices, rates, volatilities, correlations and liquidity, and reporting the loss with no probability attached. It complements VaR and ES by describing events the estimation window never contained (38, 40).

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Stressed expected shortfall (SES) The stress-scenario capital for non-modellable risk factors, calibrated factor by factor and added to the modeled charge (39).

Stressed VaR VaR computed on a fixed historical period of significant stress rather than on the recent window, so that capital cannot fall merely because markets have been calm. Added by Basel 2.5 in 2009 alongside the existing VaR; the current framework keeps the idea by calibrating expected shortfall to a period of stress (38, 40).

Strike price The price at which an option's holder may buy (a call) or sell (a put) the underlying (1, 10).

Strip A set of contracts of the same type spanning consecutive dates, or a run of strikes, traded and priced as a single position (6).

Structured note A bond whose coupon or redemption is linked to a derivative payoff (35).

Student's t distribution A bell-shaped distribution with ν degrees of freedom whose density falls off as a power, $|x|^{-(\nu+1)}$, rather than exponentially; the fewer the degrees of freedom, the fatter the tails, and it approaches the normal as ν grows. Its variance is $\nu/(\nu-2)$, so it is rescaled to a chosen standard deviation before use. With $\nu = 4$ it is the book's standard fat-tailed alternative to the normal (38).

Subadditivity The property $\rho(A+B) \le \rho(A) + \rho(B)$: diversification cannot raise measured risk. It is what lets desk limits add up to a firm limit; expected shortfall always has it and VaR does not (38, 40).

Super-senior tranche The top layer of a securitization's capital structure, above even the AAA-rated senior notes. Banks retained it as unsellable and apparently riskless, which is how the risk came back to them in 2007–08 (26).

Survival probability The probability $Q(t)$ that a borrower is still performing at t, obtained from the hazard rate as $Q(T) = \exp\bigl(-\int_0^T \lambda(u)\,du\bigr)$. A curve rather than a number; both legs of a CDS are integrals over its risk-neutral version (24, 25).

Swap An agreement to exchange cash flows on a schedule (7).

Swap market model A model that makes the forward swap rate lognormal under the annuity measure, in which it is a martingale, so that Black's formula prices a swaption exactly, as the LIBOR market model prices a caplet (32).

Swap measure The measure whose numeraire is a swap's annuity, also called the annuity measure. The forward swap rate is a martingale under it, which justifies Black's formula for swaptions; valuing a CMS payment requires a change away from it (28, 30).

Swap spread The swap rate minus the yield on a government bond of the same maturity. Long-dated dollar swap spreads have frequently been negative since 2008, which prices the balance sheet a dealer needs to hold the bond, not a view that banks are better credits than the sovereign (7).

Swaption An option to enter a swap (29).

Swing contract A fixed-price supply agreement in which the buyer chooses each day how much to take, within daily and total volume limits; the total limit links the days, so it is valued by dynamic programming over price and remaining volume (36).

Synthetic short A bought put and a written call at the same strike and expiration, which gains and loses exactly as a short stock position does; its price reflects the cost of borrowing the stock (10, 11). Also short exposure taken through a total return swap, where the dealer borrows the stock to hedge and the borrow cost dominates the spread (33).

Systematic risk Risk that moves with the market as a whole and cannot be diversified away. An asset with none has a futures price equal to the expected spot price; most assets have some,

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so the futures price sits below it (5).

T

Tail dependence The tendency of extreme outcomes to arrive together, measured as the limiting probability that one variable is extreme given that another is. The Gaussian copula has none, which is empirically false for credit (26, 38).

Tailing the hedge Reducing a futures hedge by the discount factor over the remaining hedge life, because daily settlement lets gains earn interest until the hedge is lifted. Small for short horizons and growing with maturity and the level of rates (3).

Target redemption note (TARN) A structured note that terminates once its cumulative coupons reach a stated threshold, so the redemption date itself is a function of the path and not of any single date's state (32).

Tenor The length of the period a rate applies to (overnight, one month, three months), as distinct from the maturity of the instrument that pays it. A ten-year swap whose floating rate resets every three months has a maturity of ten years and a tenor of three months (4, 7). For a swaption, the length of the underlying swap, so a 5×10 is a five-year option on a ten-year swap (29).

Term rate A forward-looking rate for a period, known at its start, which was LIBOR's defining feature. Term SOFR is constructed from SOFR futures prices rather than from bank submissions (4).

Term sheet The document setting out a structured note's terms for the buyer. It states the payoff but not that the investor is short volatility, or how often the bad outcomes occur (35).

Term SOFR A forward-looking SOFR term rate, published by CME for one, three, six, and twelve months and derived from SOFR futures prices rather than from bank submissions. Recommended by the ARRC for business loans and the derivatives hedging them, but not for the general derivatives market (4, 7, 30).

Term structure The dependence of a rate or a volatility on maturity (4, 19).

Theta The rate at which an option loses value with the passage of time (18).

Thomas algorithm Gaussian elimination specialized to a tridiagonal system, a forward sweep and a back substitution that touch only the three bands; how the implicit and Crank–Nicolson schemes are solved (20).

Tick size The smallest permitted change in a futures contract's price (2).

Time-homogeneous Of a volatility specification: depending only on the time left to maturity and not on calendar time, so the structure it implies for a year from now matches today's. Preferred for stability rather than for fit (32).

Time value An option's value in excess of its intrinsic value (10).

Title transfer / security interest The two legal forms of collateral. Under title transfer, collateral becomes the recipient's property, to be returned in equivalent form and free to be rehypothecated; under a security interest the poster keeps ownership and the recipient holds a charge it can enforce only on default (8).

Total return swap A swap exchanging an asset's total return for a financing rate; the instrument through which leverage accumulates without appearing in ownership disclosures (7, 27, 33).

Total shareholder return (TSR) Share price appreciation plus dividends. A relative TSR target is a market condition: it is built into an award's grant-date fair value, usually by simulation, and expensed even if the award pays nothing (17, 20).

Trading book / banking book The two capital regimes for a bank's positions: the trading book for positions held for sale or market making, capitalized against price moves, and the

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banking book for positions held to maturity, capitalized against default. The boundary between them is now prescriptive (39).

Tranche A layer of a securitization's loss distribution. Economically a call spread on the pool's loss (24, 25, 26).

Transition matrix The probabilities of moving from each rating to each other rating, default included, over a fixed period, usually a year. Its generator matrix gives the matrix for any other horizon (24).

Trinomial tree A tree in which the price can move up, down or stay level at each step. It absorbs state-dependent volatility into its probabilities, and the explicit finite difference scheme is one in disguise (20, 31).

V

Undertakings for Collective Investment in Transferable Securities (UCITS) The European Union's framework for funds sold to the public. A UCITS fund that measures its derivatives exposure by VaR must keep its 99%, twenty-day VaR below 20% of net asset value, or below twice that of a reference portfolio (38).

Value at risk (VaR) A loss threshold exceeded with a stated probability over a stated horizon. Not coherent, and not a worst case (38).

Vanna, volga Second-order sensitivities: vanna to spot and volatility together, how delta moves with volatility; volga, also called vomma, to volatility twice, how vega moves with volatility. What a delta-vega hedge leaves behind (18).

Variable annuity An investment-linked savings policy carrying a guaranteed minimum benefit. The guarantee depends jointly on equity and interest rate paths (32).

Variance swap A contract paying realized variance against a fixed rate; replicable from a strip of options (22).

Variation margin Collateral covering the current mark-to-market, settled as it changes (2, 8).

Variation margin gains haircutting A CCP recovery tool that withholds part of the variation margin gains owed to members whose positions have profited, taking money from the winners to pay for the losers once the default waterfall is exhausted (2).

Vega The sensitivity of an option's value to volatility (18).

Vega notional The way a variance swap is quoted: the slope of its payoff in realized volatility at the strike, so that a one-point move earns the notional to first order; variance notional is vega notional over twice the strike (22).

Vertical spread A spread whose legs share an expiration and differ only in strike, such as a bull call spread or a bear put spread. With the outright option and the risk reversal, one of the few strategies whose dominant exposure is direction (12).

VIX The 30-day variance swap rate on the S&P 500, expressed as a volatility and computed by Cboe from a discrete strip of listed options selected by a fixed rule. A well-defined number that is not the price of anything: no portfolio replicates it (19, 22).

Volatility drag The gap of $\sigma^2/2$ between the growth rate of a lognormal stock's expected value and the growth rate of its median, the return realized on a typical path (14).

Volatility cube Swaption implied volatility as a function of option expiration, underlying swap tenor and strike. Fixing the strike at the money gives a volatility surface; fixing expiration and tenor gives a smile, to which SABR is fitted (29).

Volatility risk premium The amount by which implied volatility tends to exceed the volatility later realized: typically two to four volatility points on equity indices, and more at low strikes. Partly compensation for selling protection that pays out exactly when markets fall (15, 19, 22).

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Volatility-scaled historical simulation Historical simulation in which each past return is rescaled by the ratio of today's volatility estimate to the estimate on the day it occurred, so the scenarios keep their historical shape and take today's scale; due to Hull and White. Unlike reweighting, it can produce a loss larger than any in the window (38).

Volatility surface Implied volatility as a function of strike and maturity. A quoting device that constrains, but does not determine, exotic prices (19).

Volatility swap A contract paying realized volatility, rather than variance, against a strike. Not statically replicable; priced off the variance swap less a convexity adjustment that grows with the volatility of volatility (22).

W

Warehouse A credit line from the arranging bank in which a CLO manager accumulates loans before the deal closes, with the future equity investors bearing the price risk (26).

Warrant An option issued by a company on its own stock. Exercise creates new shares, so the payoff is diluted (10, 17).

Waterfall The order in which a securitization's cash is paid out and its losses absorbed, the two running in opposite directions (26). Also the default waterfall, the sequence in which a CCP uses its default resources (2).

Wiener process A process whose increments over non-overlapping intervals are independent and normal, with variance equal to the length of the interval; also called Brownian motion, and written W in this book (14).

Worst-case default rate The default rate of a large portfolio that is exceeded in only a fraction $1 - q$ of years, the confidence-q quantile of the one-factor model's large-portfolio limit (24, 39).

Worst-of note A structured note whose principal is exposed to the worst performer among several underlyings. The investor has sold a worst-of put, and the put's premium is what pays the high coupon (34, 35).

Wrong-way risk Exposure that grows precisely as the counterparty's credit deteriorates (9, 27).

X

XVA The family of valuation adjustments (CVA, DVA, FVA, and others) applied to a derivatives book (9).

Y-Z

Yield curve The relation between yield and maturity for a class of instrument (4).

Z-spread The constant spread over the curve that reprices a bond while ignoring any embedded call (24, 31).

Zero-days-to-expiration (0DTE) option An option traded on its expiration day. Since S&P 500 options began expiring every trading day, they have grown to more than half of that market's volume; with hours to live they have almost no time value and very large gamma (10, 23).

Zero rate The rate earned on an investment made today that pays nothing until a single maturity. It can be quoted at any compounding frequency; continuously compounded, $R(t,T) = -\ln P(t,T)/(T-t)$, it is the curve every model in Part VI is fitted to (4, 31).

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Appendix D
Bibliography

This is a record of where the subject came from, not only of what the chapters cite. *A paper is listed if it mattered*—if it opened a line of work, or settled a question, or was the standard treatment for a decade—whether or not the main text discusses it and whether or not its results have since been superseded. Hull and White's 1987 stochastic-volatility paper is here for that reason, as are the short-rate models the market has moved past: a reader who does not know what was tried before cannot judge what is used now.

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